AFCD Revision Notes

Drew D. Kavi

June 18, 2026

Contents

1 Introduction

A course on social choice theory — a study of how individual preferences can be combined, typically in the setting of voting.

This field is usually referred to as *Computational Social Choice*.

The bible for this field is the book *Handbook of Computational Social Choice* [1], see Chapters 1–4.

Also, Computational Social Choice (Felix Brandt, YouTube) was a basis for this course.

2

Foundations

2.0.1 Rankings

Definition 2.0.1.1. A binary relation R on a set U is:

- Complete: if $\forall x, y \in U$, xRy or yRx.
- Antisymmetric: if $\forall x, y \in U$, xRy and yRx implies $x = y$.

Notation 2.0.1.2. For $A \subset U$ and R a binary relation on U, we write $R|_A := R \cap (A \times A)$.

Definition 2.0.1.3 (Weak Ranking)**.** A weak ordering, or weak ranking on U is a binary relation on U that is reflexive, complete and transitive. (Can be thought of as a ranking on U with ties). Let $\mathcal{R}(U)$ denote the set of all weak rankings of U.

Notation 2.0.1.4. We use $\gtrsim$ to denote a weak ordering, and write:

- $a \succ b \iff a \gtrsim b \wedge \neg(b \gtrsim a)$
- $a \sim b \iff a \gtrsim b \wedge b \gtrsim a$

Definition 2.0.1.5 (Strict Ranking)**.** A linear order, or strict ranking on U is a binary relation on U that is reflexive, complete, transitive and antisymmetric. We let $\mathcal{L}(U)$ denote the set of all strict rankings on U.

Intuition 2.0.1.6. We should think of an element of $\mathcal{L}(U)$ as placing each alternative in U at a rung from best to worst with no two sharing the same rung. In comparison, we should think of an element of $\mathcal{R}(U)$ as placing element of U at rungs from best to worst, where we allow placing multiple alternatives at the same rung (they are tied).

Observation 2.0.1.7. $\mathcal{L}(U) \subseteq \mathcal{R}(U)$.

Definition 2.0.1.8. For binary relation R on U and $\emptyset \neq A \subseteq U$,

$$\mathrm{Max}(R, A) := \{a \in A : \forall b \in A, bRa \implies aRb\}$$

Proposition 2.0.1.9. *The following hold,*

- *If R is complete, then* $\mathrm{Max}(R, A) = \{a \in A : \forall b \in A, aRb\}$
- $\mathrm{Max}(R, A)$ *can be empty*

- *If* $R \in \mathcal{R}(U)$*, then* $\mathrm{Max}(R, A) \neq \emptyset$
- *If* $R \in \mathcal{L}(U)$*, then* $\mathrm{Max}(R, A)$ *is a singleton*

Proof. The proofs follow trivially,

- We show $\forall b \in A, (bRa \implies aRb) \iff \forall b \in A, aRb$.

  The $\impliedby$ direction is immediate. For $\implies$ take $b \in A$, then bRa or aRb by completeness. If bRa, then aRb by the antecedent. If aRb, then we are done.

- Consider the rock, paper, scissor relation.
- SFC $\mathrm{Max}(R, A) = \emptyset$.

  By the first item of this proposition $\mathrm{Max}(R, A) = \{a \in A : \forall b \in A, aRb\}$. Then, since $\mathrm{Max}(R, A) = \emptyset$, for every $a \in A$, there is $b \in A$ such that $\neg(aRb)$. R is reflexive so $\neg(aRb)$ implies $b \neq a$. Also by completeness, bRa. But this implies that U is infinite (we use transitivity), contradicting the finiteness of U.

- From the above item, $\mathrm{Max}(R, A) \neq \emptyset$. Take $a, b \in \mathrm{Max}(R, A)$, then aRb and bRa. By antisymmetry, $a = b$.

□

Remark 2.0.1.10. Note we used the finiteness of U, which we assume throughout.

2.0.2 Preferences

Definition 2.0.2.1 (Prefernces over Alternatives)**.** Let U be a finite set, which we call the *alternatives*. A *prefernce relation* on U is linear order on U.

Typically, we consider multiple preference relations, we usually use the notation $\succsim_i \in \mathcal{L}(U)$ to denote the prefernces of voter i.

Remark 2.0.2.2. It is not without loss of generality to assume that preferences are linear orders, it would not be unreasonable to consider preference relations as weak orders.

In this course we assume preference relations are linear orders.

Definition 2.0.2.3 (Preference Profile)**.** Let $N = \{1, \ldots, n\}$ is a finite set of *voters* (we assume $n \geq 2$).

We call $R_N = (\succsim_1, \ldots, \succsim_n) \in \mathcal{L}(U)^N$ a *preference profile* over U [1].

Definition 2.0.2.4 (Feasible Sets)**.** We notate the *feasible sets* $\mathcal{F}(U) := \mathcal{P}(U) \setminus \{\emptyset\}$.

Definition 2.0.2.5 (Aggregation Functions)**.** A *Social Choice Function (SCF)* maps a prefernce profile and a feasible set A to a non-empty subset of A. Formally, $f : \mathcal{L}(U)^N \times \mathcal{F}(U) \to \mathcal{F}(U)$, is a social choice function if $f(R_N, A) \subseteq A$ for all $R_N \in \mathcal{L}^N$ and $A \in \mathcal{F}(U)$.

A *social welfare function (SWF)* maps a preference profile to a weak ordering on U. Formally, $g : \mathcal{L}(U)^N \to \mathcal{R}(U)$.

Example 2.0.2.6 (Trivial SCF)**.** The trivial SCF is defined as $f(R_N, A) := A$ for all $R_N \in \mathcal{L}^N$ and $A \in \mathcal{F}(U)$.

[1]We use N as the superscript (instead of n) because we think of R_N as a function from $N \mapsto \mathcal{L}(U)$.

Example 2.0.2.7 (Borda's Rule)**.** Given $|U| = m$, for a voter with linear order $\succsim_i$ and an alternative $a \in U$, define the *Borda score* as

$$s_i(a) := |\{b \in U : a \succsim_i b\}| - 1$$

i.e., the number of alternatives ranked strictly below a by voter i. The top-ranked alternative gets $m - 1$ points, the bottom gets 0.

The *total Borda score* is $S(a) := \sum_{i \in N} s_i(a)$.

As a SWF: Define $a \succsim^B b \iff S(a) \geq S(b)$. This is a weak ordering on U (reflexivity, completeness, and transitivity are inherited from $\geq$ on $\mathbb{R}$).

As a SCF: Given feasible set A, return

$$f(R_N, A) := \operatorname{argmax}_{a \in A} S(a)$$

the alternatives in A with the highest total Borda score. This is non-empty (a finite set has a maximum) and clearly a subset of A.

Note that the scores are computed over all of U, not just A. One could instead recompute Borda scores restricted to A; the two versions can disagree.

This is referred to as the *Broad Borda's rule*, and the version that first restricts to the feasible set and then computes the Borda scores is called the *Narrow Borda's rule*.

§2.1 Desired Properties and Initial Theorems

Definition 2.1.0.1 (Basic Fairness Conditions)**.** Often we hope that our aggregation functions satisfy certain properties, two most fundamental are the following (defined for SCFs):

- Anonymity: the SCF is invariant under permutations of the voters. Formally, for all pairs $R_N, R'_N \in \mathcal{L}(U)^N$ such that there is a permutation $\pi : N \to N$ with $\succsim_i = \succsim'_{\pi(i)}$, and for all $A \in \mathcal{F}(U)$, we have $f(R_N, A) = f(R'_N, A)$.
- Neutrality: the SCF is well-behaved under relabelling of the alternatives. Formally, for all pairs $R_N \in \mathcal{L}(U)^N$, $R'_N \in \mathcal{L}U'$ such that there is a bijection $\sigma : U \to U'$ with $x \succsim_i y \iff \sigma(x) \succsim'_i \sigma(y)$ for all $i \in N$ and $x, y \in U$, and for all $A \in \mathcal{F}(U)$, $\sigma(f(R_N, A)) = f(R'_N, \sigma(A))$.

Definition 2.1.0.2 (Pareto Optimality)**.** Another desirable property is Pareto optimality, which requires that if all voters prefer a to b, then b should not be chosen. More specifically, given $R_N \in \mathcal{L}(U)^N$ and $x, y \in U$, we say:

- x *Pareto-dominates* y if $\forall i \in N$, $x \succ_i y$.
- y is *Pareto-dominated* if there is some $x \in U$ that Pareto-dominates y.
- x is *Pareto-optimal* if there is no $y \in U$ that Pareto-dominates x.

A SCF f satisifies *Pareto optimality* if for all $R_N \in \mathcal{L}(U)^N$ and $A \in \mathcal{F}(U)$, $f(R_N, A) \subseteq \{x \in A : x \text{ is Pareto-optimal}\}$.

One can define a *SCF* $f_{\text{Pareto}}(R_N, A) := \{x \in A : x \text{ is Pareto-optimal}\}$ that returns all Pareto-optimal alternatives.

NB1: Pareto-optimal alternatives always exist, to see as such, pick an alternative, if it is Pareto-optimal, we are done, otherwise there is an alternative that strictly Pareto-dominates it, consider this alternative, if it is

Pareto-optimal, we are done, otherwise there is an alternative that strictly Pareto-dominates it. Given that U is finite, all that remains is to show that we cannot cycle here, but this is clear since the strict Pareto-relation is transitive (and irreflexive).

NB2: Pareto-dominated alternatives need not exist, consider two voters with opposite preferences over two alternatives a and b.

Definition 2.1.0.3 (Resoluteness)**.** A SCF f is *resolute* if for all $R_N \in \mathcal{L}(U)^N$ and $A \in \mathcal{F}(U)$, $f(R_N, A)$ is a singleton.

Proposition 2.1.0.4. *In general, an anonymous and neutral SCF is not resolute.*

Proof. Counter-example, take $U = \{a, b\}$ and $N = \{1, 2\}$. Then the profile:

$$R_N = \begin{pmatrix} a & b \\ b & a \end{pmatrix}$$

forces any anonymous and neutral SCF to return $\{a, b\}$ for the feasible set $A = \{a, b\}$.

In more detail, suppose for contradiction that f is anonymous, neutral, and resolute. Then $f(R_N, A) = \{a\}$ or $\{b\}$. WLOG assume $f(R_N, A) = \{a\}$. Consider now the profile R'_N obtained by swapping the two voters, i.e.,

$$R'_N = \begin{pmatrix} b & a \\ a & b \end{pmatrix}$$

By anonymity, $f(R'_N, A) = \{a\}$. But also by neutrality, $f(R'_N, A) = \{b\}$. Contradiction. □

Theorem 2.1.0.5 (Moulin, 1983)**.** *For n votes and m alternatives, there exists an anonymous, neutral, resolute, and Pareto-optimal SCF if and only if n is not divisible by any $k \in [2, \ldots, m]$.*

In fact, for n not divisible by any $k \in [2, \ldots, m]$, then the instant runoff *social choice function is anonymous, neutral, resolute, and Pareto-optimal.*

Definition 2.1.0.6 (Strategyproofness)**.** A SCF f is *strategyproof* if voters are never better off by misrepresenting their preferences.

Formally, we extend $\succsim_i$ to singleton sets (only), so that $\{x\} \succ_i \{y\} \iff x \succ_i y$. And we say that a SCF f is *manipulable* (by voter i), if there exists $R_N, R'_N \in \mathcal{L}(U)^N$, $A \in \mathcal{F}(U)$, such that $\succsim_j = \succsim'_j$ for all $j \neq i$, and $f(R'_N, A), f(R_N, A)$ are singletons and $f(R'_N, A) \succ_i f(R_N, A)$.

A SCF is *strategyproof* if it is not manipulable.

Definition 2.1.0.7 (Monotonicty)**.** A SCF f is *monotone* if the promotion of an alternative a in some voter's preference cannot cause a to be demoted from the set of winners.

Formally, f is monotone if for all $R_N, R'_N \in \mathcal{L}(U)^N$ and $A \in \mathcal{F}(U)$, such that the following hold:

- $\succsim_j = \succsim'_j$ for all $j \neq i$
- $x \succsim y \iff x \succsim' y$ for all $x, y \in U \setminus \{a\}$
- $a \succsim_i x \implies a \succsim'_i x$ for all $x \in U \setminus \{a\}$

then if $a \in f(R_N, A)$, $a \in f(R'_N, A)$.

Moreover, we say f is *positive responsive* if for all $R_N, R'_N \in \mathcal{L}(U)^N$ and $A \in \mathcal{F}(U)$, such that the same conditions hold, and where $R_N \neq R'_N$, then if $a \in f(R_N, A)$, $\{a\} = f(R'_N, A)$.

Intuition 2.1.0.8. Clearly, positive responsiveness implies monotonicity. It is often easy to satisfy monotonicity, but much harder to satisfy positive responsiveness. For example, plurality is monotone but not positive responsive.

Proposition 2.1.0.9. *For a resolute SCF f on two alternatives, the following are equivalent.*

1. *f is strategyproof*
2. *f is monotone*
3. *f is positive responsive*

Proof.
(2) $\iff$ (3): This biconditional is trivial since resoluteness and monotonicity together imply positive responsiveness, and positive responsiveness always implies monotonicity.

(2) $\implies$ (1): As is typical in these settings, we proceed by contrapositive. Suppose f is manipulable by voter i. There exists profiles $R_N, R'_N \in \mathcal{L}(U)^N$ differing only in voter i's preferences, and $A \in \mathcal{F}(U)$ such that $f(R'_N, A) \succ_i f(R_N, A)$ — resoluteness gives that $f(R'_N, A)$ and $f(R_N, A)$ are singletons. Since $f(R'_N, A) \succ_i f(R_N, A)$ then $R'_N \neq R_N$. Without loss of generality, assume that the i-th preference relation of R_N and R'_N are

$$\succsim_i = \begin{pmatrix} a & b \end{pmatrix} \qquad\qquad \succsim'_i = \begin{pmatrix} b & a \end{pmatrix}$$

respectively. Since $f(R'_N, A) \succ_i f(R_N, A)$, $f(R'_N, A) = \{a\}$ and $f(R_N, A) = \{b\}$. It is clear that R_N and R'_N agree on all preference relations but that of voter i. It is vacuous that the i-th preference relation agrees the those relations not involving b. It is clear that R'_N strictly promotes b compared to R_N for voter i. But, $b \notin f(R'_N, A)$, so f is not monotone.

(1) $\implies$ (2): Again, by contrapositive, assume f is not monotone. Then there are profiles $R_N, R'_N \in \mathcal{L}(U)^N$ and $A \in \mathcal{F}(U)$ such that $f(R_N, A) = \{a\}$ and $f(R'_N, A) = \{b\}$, and R_N and R'_N differ only in voter i strictly promoting a in R'_N compared to R_N. As such, if voter i misrepresents their preferences as in R_N, then they get b, which they prefer to a. Thus, f is manipulable by voter i, and not strategyproof. $\square$

Remark 2.1.0.10. As mentioned, positive responsiveness is rather difficult to satisfy, and for this reason we often consider a weakening of it that only holds for feasible sets of size 2; and denote this as *positive responsiveness*$_2$ *(PR$_2$)*.

Notation 2.1.0.11. For alternatives, a, b and preference profile R_N, we write $N_{ab} := \{i \in N : a \succsim_i b\}$, and $n_{ab} := |N_{ab}|$.

Intuition 2.1.0.12. Even with anonymity, neutrality, and only two alternatives, there are still several choices of SCFs. For one, we can define a threshold percentage of the vote won to have a single winner.

We can do weird things, like picking the minority vote, or defining a function based on if n_{ab}, n_{ba} is prime or not or is odd or not.

To rule out these more obtuse SCFs, we need something like monotonicity, or strategyproofness — intuitively they give that the SCF *moves* in the right direction.

Remark 2.1.0.13. Informal arguments suffice for showing neutrality and anonymity for exams - we need not go to first principles.

§2.2 May's Theorem(s)

Theorem 2.2.0.1 (May's Theorem)**.** *For two alternatives, the only SCF that is anonymous, neutral, and positive responsive is plurality.*

Proof. Consider that for two alternatives a and b, any anonymous SCF f must be a function of n_{ab} and n_{ba} and the choice of feasible set A only (strictly speaking, we do not need n_{ba}).

Let $f(m) := f(n_{ab}(R_N), \{a,b\})$. If $f(0) = \{a\}, \{a,b\}$, then (for $m > 1$) we get $f(m) = \{a\}$, violating neutrality. Thus, (and by symmetry) $f(0) = \{b\}$ and $f(m) = \{a\}$.

By positive responsiveness, if $a \in f(k)$ then $f(l) = \{a\}$ for each $l > k$. Let $k^* := \min\{k : a \in f(k)\}$. Suppose $k^* > \lceil n/2 \rceil$, then by considering the profile with a and b swapped we contradict neutrality. Likewise, for $k^* < \lceil n/2 \rceil$ a similar argument holds.

For $k^* = \lceil n/2 \rceil$, it is clear that neutrality gives $f(k^*) = \{a, b\}$.

□

Theorem 2.2.0.2. *For two alternatives, of all the SCFs that are anonymous, neutral, and monotonic, plurality is the most* decisive *(give the fewest ties).*

Remark 2.2.0.3. It is so nice, and well-axiomatically motivated to use plurality when we have only two alternatives, and thus, a natural idea is to use plurality even when we have more than two alternatives.

§2.3 Further Properties and Impossibility results

Notation 2.3.0.1. $\succsim_M$ is the *pairwise majority relation*, defined by $a \succsim_M b \iff n_{ab} \geq n_{ba}$.

Observation 2.3.0.2 (Condorcet Paradox). $\succsim_M$ may contain cycles. The Condorcet Profile is:

$$R_N = \begin{pmatrix} a & b & c \\ b & c & a \\ c & a & b \end{pmatrix}$$

and contains cycles,

$$a \succ_M b, b \succ_M c, c \succ_M a$$

Definition 2.3.0.3 (Transitive Rationalisability)**.** A SCF f is *transitively rationalisable* if for each $R_N \in \mathcal{L}(U)^N$ there is a weak ordering R on U such that $f(R_N, A) = \text{Max}(R, A)$ for all $A \in \mathcal{F}(U)$.

Remark 2.3.0.4. The Condorcet Profile is not transitively rationalisable. Typically, in many rational decision making theories transitivity is thought to be a cornerstone. The Condorcet Paradox shows that despite individuals making transitive choices, the group choice can be intransitive.

Theorem 2.3.0.5. *The Condorcet Paradox can be alternatively be phrased as an impossibility result:*

> *There does not exists a SCF that is anonymous, neutral, positive responsive$_2$, and transitively rationalisable.*

Sketch. The first three properties essentially force pairwise choices according to majority rule. But then a Condorcet Profile can be constructed, which, given we make pairwise choices according to majority rule, is not transitively rationalisable. □

Definition 2.3.0.6. A Condorcet Winner is an alternative a such that $a \succ_M b$ for all $b \neq a$. Clearly, if a Condorcet winner exists, it does uniquely.

Definition 2.3.0.7 (Independence of Infeasible Alternatives (IIA))**.** A SCF f satisfies (IIA) if, for all A, R_N, R'_N: $R_N|_A = R'_N|_A$ implies $f(R_N, A) = f(R'_N, A)$.

Example 2.3.0.8. Consider the following pair of profiles,

R_N :

z_1	z_2	z_3
a	y	y
b	a	x
c	c	z
x	b	b
y	x	a
z	z	c

R'_N :

z_1	z_2	z_3
x	y	c
c	x	b
y	a	a
z	b	y
b	z	x
a	c	z

When restricted to $A := \{x, y, z\}$, they are identical. If f satisfies IIA, then $f(R_N, A) = f(R'_N, A)$.

Observation 2.3.0.9. A natural question in the set-up, is: Do we really need to know all the preferences of voters over all alternatives to determine the winner? If the SCF satisfies IIA then we do not.

Theorem 2.3.0.10 (Arrow's Impossibility Theorem)**.** *There is no SCF that satisfies IIA, Pareto-Optimality, non-dictatorship, and transitive rationalisability when there are at least three alternatives and at least two voters.*

An alternative formulation is *every SCF that satisfies IIA, Pareto-optimality, and transitive rationalisability is a* dictatorship [2].

Most commonly, Arrow's Impossibility Theorem is stated for SWFs. To be precise, we should define the notions of IIA, Pareto-optimality, transitive rationalisability, and dictatorship for SWFs.

Definition 2.3.0.11 (Properties for SWFs)**.** Let g be a SWF. For $R_N, R'_N \in \mathcal{L}(U)^N$, let $\gtrsim := g(R_N)$ and $\gtrsim' := g(R'_N)$.

- g satisfies *independence of irrelevant alternative (IIA)* if for all R_N, R'_N, x, y, $R_N|_{\{x,y\}} = R'_N|_{\{x,y\}}$ implies $\gtrsim |_{\{x,y\}} \iff \gtrsim' |_{\{x,y\}}$.
- g satisfies *Pareto optimality* if for all R_N and x, y, $\forall i \in N, x \succ_i y$ implies $x \succ y$.
- g is dictatorial if if there is some $i \in N$ such that for all R_N, $g(R_N) = \gtrsim_i$.

Remark 2.3.0.12. Note that in a sense the notion of dictatorship for SWFs is stronger than for SCFs since the dictator can dictate not just the winners but also the entire ordering output.

[2]Where by a dictatorship we mean that the winner is always the top ranked alternative of a specific voter.

Below, we drop the transitive rationalisability condition, since that is baked into the notion of a SWF.

Theorem 2.3.0.13 (Arrow's for SWFs)**.** *Every SWF that satisfies IIA and Pareto optimality is a dictatorship for three or more alternatives (and two or more voters which we now assume from here on out).*

Proofs of these formulations of Arrow's Impossibility Theorem are omitted, but we sketch the idea for two voters and three alternatives in the SWF case. The following table gives in each cell a preference profile, the row represents the preferences of voter 1 and the column represents the preferences of voter 2.

We fill out the table using just Pareto-Optimality for now, clearly along the diagonal any Pareto-Optimal SWF gives the same ordering as the voters, and elsewhere we can at least denote that one alternative must strictly beat another one.

	abc	acb	bac	bca	cab	cba
abc	abc	ab, ac	ac, bc	bc	ab	
acb	ab, ac	acb	ac		ab, cb	cb
bac	ac, bc	ac	bac	bc, ba		ba
bca	bc		bc, ba	bca	ca	ba, ca
cab	ab	ab, cb		ca	cab	ca, cb
cba		cb	ba	ba, ca	ca, cb	cba

Now, consider the cases in the table in more detail. Consider the cell corresponding to row (a, b, c) and column (b, c, a). We can do case analysis here. First suppose that g gives $a \succ b$ in this cell. Then, by IIA, any cell where the row heading prefers a to b and the column heading prefers b to a must have $a \succ b$.

This, with transitivity fills out more of the table. We repeatedly apply this reasoning to some other case. Eventually, we see that we get a dictatorship, and this holds no matter what case we choose.

Remark 2.3.0.14. What now? Arrow's and variations on this theorem are prevalent, and so if we want these seemingly easy and desirable properties it seems we must give up. We spend much of the course finding ways to escape from this impossibility:

- restricted domains of preferences
- replace rationalisability with a weaker condition (variable electorate conditions)
- weaken rationalisability and/or IIA

3

Domain Restrictions

Remark 3.0.0.1. Implicit in Arrow's Impossibility Theorem is the assumption that all preference profiles are possible, in the literature this assumption is called *universal domain*. In some contexts, it makes sense to restrict the domain of possible preference profiles, which allows for better axiomatic results.

Example 3.0.0.2. Trivial Example: Only preference profiles are those in which all voters vote identically.

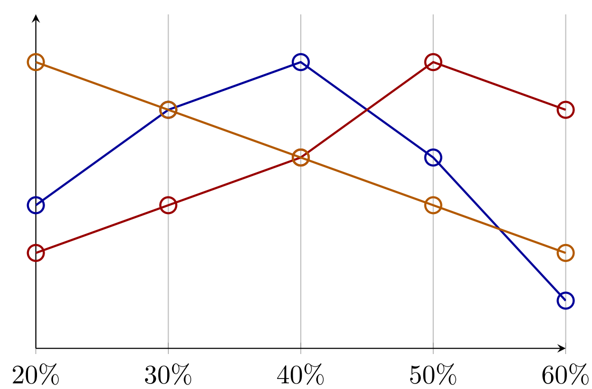
Single-peaked preferences are one of the most useful domain restrictions we can make. The following gives an example of what a single peaked preference is.

Example 3.0.0.3. Consider preferences over the maximal income tax rate. We have alternatives

$$U = \{20\%, 30\%, 40\%, 50\%, 60\%\}$$

and voters Alice, Bob, and Charlie with the following preferences:

○ Alice	○ Bob	○ Charlie
40%	50%	20%
30%	60%	30%
50%	40%	40%
20%	30%	50%
60%	20%	60%



Here, as is a reasonable assumption, the preferences of voters is single-peaked in some sense, according to the ordering of preference in the natural way. Of course, there is not usually a natural way to order preferences. We see more on single-peaked preferences later.

Definition 3.0.0.4. A restricted domain $\mathcal{D}(U)$ is a subset of $\mathcal{L}(U)^N$.

Example 3.0.0.5.

- The universal domain $\mathcal{D}_{\text{Univ}}(U) := \mathcal{L}(U)^N$.

- $\mathcal{D}_{\text{ab}}(U) = \{R_N \in \mathcal{L}(U)^N : (\forall i \in N, a \succ_i b) \vee (\forall i \in N, b \succ_i a)\}$ is the domain where all voters agree on the relative ranking of a and b.

Definition 3.0.0.6. A domain restriction $\mathcal{D}(U)$ is *Cartesian* if $\mathcal{D}(U) = \mathcal{L}'(U)^N$ for some $\mathcal{L}'(U) \subseteq \mathcal{L}(U)$.

In a Cartesian domain, the admissible preferences of a voter are independent of the preferences of other voters (which need not be true).

§3.1 Transitive Domains

Remark 3.1.0.1. $\mathcal{D}_{\text{ab}}(U)$ is not Cartesian, but $\mathcal{D}_{\text{Univ}}(U)$ is Cartesian.

Remark 3.1.0.2. As discussed, we use restricted domains to bypass impossibility results. In particular, we are interested in domain in which the majority relation $\succsim_M$ is transitive. It is clear that in this restricted domain, $\succ_M$ contains no cycles, and so we get rid of the Condorcet Paradox.

In particular, if $\succsim_M$ is transitive then $\succsim_M \in \mathcal{R}(U)$, thus $\text{Max}(\succsim_M, A)$ is a well-defined SCF.

Comment. We often like to think of (U, $\succsim_M$) as a graph.

Definition 3.1.0.3. Recall that an alternative $a \in U$ is a Condorcet winner w.r.t preference profile R_N if $a \succ_M b$ for all $b \neq a$.

We say that $a \in U$ is a weak Condorcet winner w.r.t to R_N if $a \succsim_M b$ for all $b \neq a$.

Proposition 3.1.0.4. *If $\succsim_M$ is transitive, then there exists a weak Condorcet winner.*

Proof. A transitive and complete relation on finite set has a maximum element. □

Remark 3.1.0.5. We thus, motivate the domain restricted to just those profiles with transitive majority relation, since we have a natural SCF that just outputs the weak Condorcet winners. This, couldn't be done in the general case since the Condorcet Paradox shows that we may have no weak Condorcet winners.

Definition 3.1.0.6. Let $\mathcal{D}_{\text{TRANS}}(U) := \{R_N \in \mathcal{L}(U)^N : \succsim_M \text{ is transitive}\}$.

As demonstrated if $\succsim_M$ is transitive then $\text{Max}(\succsim_M, A)$ is a well-defined SCF, and its elements are exactly the weak Condorcet winners, since they win or tie all pairwise majority comparisons.

Remark 3.1.0.7. $\mathcal{D}_{\text{TRANS}}(U)$ is not Cartesian.

Remark 3.1.0.8. In $\mathcal{D}_{\text{TRANS}}(U)$, the SCF $\text{Max}(\succsim_M, A)$ satisfies all the conditions of Arrow's Theorem.

It should be clear that it is not a dictatorship.

Consider that if everyone ranks one alternative over the other, then the majority must strictly (in fact trivially) rank that alternative over the other, and so clearly $\text{Max}(\succsim_M, A)$ satisfies Pareto optimality.

Consider that if A is a feasible set and R_N, R'_N are two profiles that are the same when restricted to A. Then consider that the pairwise majority in R_N and R'_N must be the same when restricted to A (so IIA is satisfied).

Finally, it is obvious that the majority relation is a transitive rationalisation of $\text{Max}(\succsim_M, A)$.

Lemma 3.1.0.9. $\text{Max}(\succsim_M, A)$ *is strategyproof in $\mathcal{D}_{TRANS}(U)$.*

Proof. Suppose for contradiction it is not. Then suppose that we have two profiles R_N, R'_N that differ only in voter i's preferences, and $A \in \mathcal{F}(U)$ such that $\text{Max}(\succsim'_M, A) \succ_i \text{Max}(\succsim_M, A)$ where $\text{Max}(\succsim'_M, A), \text{Max}(\succsim_M, A)$ are singletons. That is there is only one weak condorcet winner.

Let a be the weak Condorcet winner in R_N and a' be the weak Condorcet winner in R'_N. Thus, $a \succ_M a'$, $a' \succ'_M a$ (strictly because otherwise a' (resp. a) would also be weak condorcet winners, contradicting uniqueness), and $a' \succ_i a$. But then, consider that since $a \succ_M a'$, and $a' \succ_i a$, and R_N and R'_N differ only in voter i's preferences, the relative ranking in i's preferences between a and a' can only keep the same relative ranking or promote a compared to a'. Thus, it must be that $a \succ'_M a'$. Contradiction.

□

Corollary 3.1.0.10. *If $\mathcal{D}(U)$ is a subdomain of $\mathcal{D}_{TRANS}(U)$ (i.e. $\mathcal{D}(U) \subseteq \mathcal{D}_{TRANS}(U)$) then clearly the same proof above follows, and so we get that* $\text{Max}(\gtrsim_M, A)$ *is strategyproof in $\mathcal{D}(U)$.*

Remark 3.1.0.11. If there are an odd number of voters, then $\text{Max}(\gtrsim_M, A)$ is resolute in $\mathcal{D}_{\text{TRANS}}(U)$.

Proof. Clear. Cannot be that $n_{ab} = n_{ba}$. □

Theorem 3.1.0.12 (Cambell and Kelly, 2003)**.** *If n is odd,* $\text{Max}(\gtrsim_M, A)$ *is the only SCF that is non-dictatorial, Pareto-optimal, strategyproof, and resolute in $\mathcal{D}_{TRANS}(U)$.*

Proof. Omitted. □

§3.2 Single-Peaked Domains

We see that the domain $\mathcal{D}_{\text{TRANS}}(U)$ is a nice domain to work with, but perhaps it feels like a contrived example, we now turn to a natural example of a domain that is a subdomain of $\mathcal{D}_{\text{TRANS}}(U)$.

Definition 3.2.0.1. Let $\trianglelefteq \in \mathcal{L}(U)$ be a linear order of alternative. A preference relation $\gtrsim \in \mathcal{L}(U)$ is *single-peaked with respect to* $\trianglelefteq$ if for all $x, y, z \in U$: ((x $\triangleleft$ y $\triangleleft$ z) or (z $\triangleleft$ y $\triangleleft$ x)) implies ($x \succ y \implies y \succ z$).

This says, that if the linear order ranks y between x and z, then if the preference relation ranks x over y, then it must rank y over z. This gives a kind of monotonicity condition which is the intuitive notion of single-peakedness.

Alternatively, consider what this definition forbids. That is, it disallows the middle alternative from being ranked lowest (think this over).

Definition 3.2.0.2. For a linear order $\trianglelefteq \in \mathcal{L}(U)$, and a preference profile $R_N \in \mathcal{L}(U)^N$, we say that R_N is *single-peaked with respect to* $\trianglelefteq$ if each preference relation in R_N is single-peaked with respect to $\trianglelefteq$.

Definition 3.2.0.3. We define the *domain of single-peaked preferences* with respect to $\trianglelefteq$ as

$$\mathcal{D}_{\text{SP}}^{\trianglelefteq}(U) := \left\{R_N \in \mathcal{L}(U)^N : R_N \text{ is single-peaked with respect to } \trianglelefteq\right\}$$

Equivalently, we can define it as

$$\mathcal{D}_{\text{SP}}^{\trianglelefteq}(U) := \{\gtrsim \in \mathcal{L}(U) : \gtrsim \text{ is single-peaked with respect to } \trianglelefteq\}^N$$

Clearly, under this definition we see that $\mathcal{D}_{\text{SP}}^{\trianglelefteq}(U)$ is Cartesian.

Remark 3.2.0.4. Single-peaked arises naturally in a few situations,

- The desired distance of a train station from your house
- Left-Right political spectrum

- Age for people to be allowed to vote

The big motivation for considering single-peaked preferences, is the following theorem, which follows(-ish) from the previous section.

Theorem 3.2.0.5 (Black, 1948)**.** *$\succsim_M$ is transitive in $\mathcal{D}^{\trianglelefteq}_{SP}(U)$.*

- *In particular, if n is odd, every single-peaked profile admits a (strict) Condorcet winner.*
- *For even n, weak Condorcet winners are guaranteed*

We defer the proof to ???, from which it follows trivially.

Remark 3.2.0.6. The above also holds for so-called, *single-caved* preferences $\mathcal{D}^{\trianglelefteq}_{\mathrm{SC}}(U)$. A preference relation $\succsim$ is single-caved with respect to $\trianglelefteq$ if for all $x, y, z \in U$: $((\mathrm{x} \triangleleft \mathrm{y} \triangleleft \mathrm{z})$ or $(\mathrm{z} \triangleleft \mathrm{y} \triangleleft \mathrm{x}))$ implies $(y \succ x \implies z \succ y)$.

Intuition 3.2.0.7. We give some intuition as to why $\succsim_M$ is transitive in $\mathcal{D}^{\trianglelefteq}_{\mathrm{SP}}(U)$.

Assume that n is odd, let t_i denote the alternative ranked highest by voter i (the *top-choice* of voter i). We can imagine placing pebbles along the linear order $\trianglelefteq$ for each voter according to their top choice. Then the alternative that the median pebble (according to $\trianglelefteq$) is placed on is the Condorcet winner.

Sketch. See lectures for a sketch of the proof.

□

Remark 3.2.0.8. This proves something that is just a slightly weaker Black's theorem.

Remark 3.2.0.9 (Median Voting). By the above, that just the voters' top choices are sufficient to compute the Condorcet winner (when one exists).

> Importantly, we note that the plurality winner may differ from the Condorcet winner, even in $\mathcal{D}^{\trianglelefteq}_{\mathrm{SP}}(U)$.

In the domain $\mathcal{D}^{\trianglelefteq}_{\mathrm{SP}}(U)$ this — which is just $\mathrm{Max}(\succsim_M, A)$ — is known as *median voting.*

Median voting satisfies strategyproofness in the domain $\mathcal{D}^{\trianglelefteq}_{\mathrm{SP}}(U)$. This follows from Black's theorem and Lemma 3.1.0.9.

We see that in $\mathcal{D}^{\trianglelefteq}_{\mathrm{SP}}(U)$, it is computationally trivial to find the Condorcet winner. So far, we have not thought much about the computational social choice theory from a complexity perspective, but this is a theme of the course that we will now be seeing more of.

Remark 3.2.0.10. Can we efficiently check whether a preference profile is single-peaked with respect? If we are given some order $\trianglelefteq$, we can trivially check whether a preference profile is single-peaked with respect to $\trianglelefteq$ by verifying the single-peaked condition for each voter's preferences We check the single-peaked condition for each preference relation in a profile by going through each voter's contiguous 3-tuples of preferences (think!) i.e. in time linear in the number of alternatives. Thus, given $\trianglelefteq$, we can check whether a profile is single-peaked with respect to $\trianglelefteq$ in time $O(n|U|)$.

The more interesting question is whether we can efficiently check single-peakedness without being given $\trianglelefteq$; after all, we have all these nice properties if a profile is single-peaked with respect to some order, so it is nice to be able to easily check whether a profile is single-peaked. Of course, checking all linear orders is not efficient.

Observation 3.2.0.11. A least-ranked alternative in a preference relation must be placed at the leftmost or rightmost end of the linear order.

Reason why this is true?

Thus, there can be at most two least-ranked alternatives. This motivates the idea that we try to arrange the alternatives from the outside to the inside.

Idea 3.2.0.12. From this, we attempt to construct a linear order by placing the least-ranked alternatives at the ends, and iterating this. Suppose that after one iteration l and r are the least-ranked alternatives in the profile and so they get placed at the innermost left and right positions of the partially constructed linear order. Then, we remove l and r from the profile and repeat this process. Let $A \subseteq U$ be the set of alternatives that have not yet been arranged.

Suppose that x is a bottom ranked alternative in the profile restricted to A (so needs to be placed next). If there is a voter that ranks $l \succ_i x \succ_i r$, and that voter does not rank x as their top choice when we restrict to A, then we have to place x on the right (next to r).

Why?

Suppose for contradiction that x is placed on the left, just inside l. Then in $\trianglelefteq$ the arrangement is:

$$\cdots \quad l \quad x \quad \underbrace{A \setminus \{x\}}_{\ni\, t_i} \quad r \quad \cdots$$

where t_i denotes the peak of voter i restricted to A, which is not x by assumption. Since $t_i \in A \setminus \{x\}$, it lies strictly to the right of x in $\trianglelefteq$. Moving leftward from t_i, we reach x before l, so x is closer to t_i than l is. Single-peakedness then requires $x \succ_i l$. But voter i has $l \succ_i x$, a contradiction. Thus x must be placed on the right, next to r.

Check this makes sense.

Algorithm 1 Single-Peaked Recognition

Set dummy leftmost alternative to $z_l \notin U$, rightmost to $z_r \notin U$
$l \leftarrow z_l$, $r \leftarrow z_r$
$A \leftarrow U$
while $|A| \geq 2$ **do**
  $B \leftarrow \{x \in A : x$ is bottom-ranked in $R_N|_A$ for some voter$\}$
  $R \leftarrow \{x \in B : \exists i \in N$ such that $l \succ_i x \succ_i r$ and $t_i(A) \neq x\}$
  $L \leftarrow \{x \in B : \exists i \in N$ such that $r \succ_i x \succ_i l$ and $t_i(A) \neq x\}$
  if $|B| \leq 2 \wedge |L| \leq 1 \wedge |R| \leq 1 \wedge L \cap R = \emptyset$ **then**
    Place the alternative in L (if any) next to l; update l
    Place the alternative in R (if any) next to r; update r
    Place any $x \in B \setminus (L \cup R)$ in an arbitrary empty slot next to l or r
  else
    return R_N is not single-peaked
  end if
  $A \leftarrow A \setminus B$
end while
if $|A| = 1$ **then**
  Place the remaining $x \in A$ in the last slot
end if
return R_N is single-peaked with the constructed order $\trianglelefteq$

TODO: Write and understand this in more detail, also trace this algorithm for an example.

Theorem 3.2.0.13. *The above algorithm is correct and runs in time $O(n|U|)$ — so in time that is linear in in the size of a specification of a preference profile.*

Proof. See [2][Section 4.1.]. □

Remark 3.2.0.14. Essentially the same algorithm can be used to check whether a profile is single-caved, by running it on a preference profile that is flipped upside down.

Remark 3.2.0.15. We have noted that $\succsim_M$ is transitive in $\mathcal{D}_{\mathrm{SP}}^{\trianglelefteq}(U)$ and in $\mathcal{D}_{\mathrm{SC}}^{\trianglelefteq}(U)$. Therefore, it is too in the union domain,

$$\mathcal{D}_{\mathrm{SP}}^{\trianglelefteq}(U) \cup \mathcal{D}_{\mathrm{SC}}^{\trianglelefteq}(U)$$

One might ask if this is maximal, that is, do there exist profiles that are neither single-peaked nor single-caved, which still have a transitive majority relation? The answer is no, consider the following example.

1	**1**	**1**
b	a	d
c	b	b
a	c	c
d	d	a

That is because, the above profile satisfies the property of *single-peakedness for triples* — defined in a rather obvious way.

Can we characterise the (Cartesian) domains for which the majority relation is transitive [1]?

Definition 3.2.0.16 (Value-Restricted Domain)**.** A Cartesian domain $\mathcal{D}(U) = \mathcal{L}'(U)^N$ with $\mathcal{L}'(U) \subseteq \mathcal{L}(U)$ is *value-restricted* if for each $x, y, z \in U$, there is some alternative, say x, such that

- x is never the worst alternative ($\forall \succsim \in \mathcal{L}'(U) : x \succ y \vee x \succ z$), or
- x is never the best alternative ($\forall \succsim \in \mathcal{L}'(U) : y \succ x \vee z \succ x$), or
- x is never the middle alternative ($\forall \succsim \in \mathcal{L}'(U) : (x \succ y \wedge x \succ z) \vee (y \succ x \wedge z \succ x)$).

The forbidden substructure for value-restricted domains is the Condorcet profile restricted to any triple:

x	y	z
y	z	x
z	x	y

$\mathcal{D}_{\mathrm{SP}}^{\trianglelefteq}(U)$ and $\mathcal{D}_{\mathrm{SC}}^{\trianglelefteq}(U)$ are value-restricted for all $\trianglelefteq$.

Theorem 3.2.0.17 (Sen and Pattanaik, 1969)**.** *Let $\mathcal{L}'(U) \subseteq \mathcal{L}(U)$ and n odd. Then, $\succsim_M$ is transitive in domain $\mathcal{D}(U) = \mathcal{L}'(U)^N$ iff $\mathcal{D}(U)$ is value-restricted.*

[1]We restrict to Cartesian domains, otherwise by definition it is just $\mathcal{D}_{\mathrm{TRANS}}$.

Remark 3.2.0.18. This does **not** imply that every preference profile for which $\succsim_M$ is transitive, is value-restricted. For example, consider the following profile:

3	**1**	**1**
a	b	c
b	c	a
c	a	b

Here $\succsim_M$ is transitive (since a wins every pairwise comparison), yet $\mathcal{L}'(U) = \mathcal{L}(U)$ contains the full Condorcet triple, so the domain is not value-restricted.

4

Borda, Condorcet, and Kemeny

Idea 4.0.0.1. The *theme* of this chapter comes from the conflict between the social choice ideas of Borda and Condorcet.

Definition 4.0.0.2. In this chapter, we consider a variable set of voters.

Formally speaking, we consider the set of preference profiles over $\bigcup_{N \subseteq \mathbb{N}; |N| < \infty} \mathcal{L}(U)^N$. This allows us to consider profiles with different numbers of voters, and consider merging profiles $R_N \in \mathcal{L}(U)^N$ and $R_{N'} \in \mathcal{L}(U)^{N'}$ into a profile $R_N \cup R_{N'} \in \mathcal{L}(U)^{N \cup N'}$ (assuming $N \cap N' = \emptyset$).

Definition 4.0.0.3 (Scoring Rule)**.** Informally speaking, for a fixed feasible set $A \subseteq U$, a *score vector* $s := (s_1, s_2, \ldots, s_{|A|})$ is a real vector, such that if voter ranks an alternative at the i-th position, then it gets s_i points. A *scoring rule* is a SCF that chooses those alternatives with the highest total score.

More formally, we should define the notion of a score vector for each possible size of feasible set. Let $s^{(1)}, s^{(2)}, \ldots, s^{(|U|)}$ be a collection of score vectors where $s^{(k)}$ has size k. The corresponding scoring rule is defined as $\operatorname{argmax}_{x \in A} s(x, A)$, where,

$$s(x, A) := \sum_{i \in N} s^{(|A|)}_{|y \in A : y \succsim_i x|}$$

Remark 4.0.0.4. The above definition is a bit weird/annoying, it allows for perversities like Borda for even sized feasible sets and Plurality for odd sized feasible sets. Usually, we think about the feasible set as fixed and often omit it from the notation.

Example 4.0.0.5.

- Borda's rule, $s^{(|A|)} := (|A| - 1, |A| - 2, \ldots, 0)$.

  Think if this is broad or narrow Borda?
- Plurality, $s^{(|A|)} := (1, 0, \ldots, 0)$
- *Anti-Plurality*, $s^{(|A|)} := (1, 1, \ldots, 1, 0)$

Proposition 4.0.0.6. *Scoring rules are invariant under* positive affine transformations [1] *of their score vectors.*

Proof. Easy boring exercise. ∎

[1] $s_i^{(k)} \mapsto \alpha_i s_i^{(k)} + \beta_i$ where $\alpha_i > 0$.

Proposition 4.0.0.7. *Scoring rules can be efficiently computed* [2].

Proof. We compute them in the obvious way. □

Proposition 4.0.0.8. *A scoring rule is monotonic if and only if for all k, $s_1^{(k)} \geq s_2^{(k)} \geq \ldots \geq s_k^{(k)}$.*

A monotonic scoring rule is non-trivial *if $s_1^{(k)} > s_k^{(k)}$.*

Proof. Exercise. □

Definition 4.0.0.9 (Reinforcement)**.** A SCF f satisfies *reinforcement* if for all A and disjoint sets of voters N and N', and all $R_N \in \mathcal{L}(U)^N$ and $R_{N'} \in \mathcal{L}(U)^{N'}$, such that $f(R_N) \cap f(R_{N'}) \neq \emptyset$,

$$f(R_N \cup R_{N'}, A) = f(R_N, A) \cap f(R_{N'}, A)$$

Informally, whenever there is a common alternative selected by two disjoint sets of voters (under the SCF), then the common alternatives should be the only alternatives selected when we merge the two sets of voters together.

Example 4.0.0.10. Let's think about which rules satisfy this. It is clear that Plurality satisfies this.

Think why? Also, does Borda satisfy it?

Theorem 4.0.0.11 (Smith, 1973; Young, 1975)**.** *A neutral and anonymous SCF satisfies reinforcement if and only if it is a* composed *scoring rule.*

Definition 4.0.0.12 (Composed scoring rule)**.** An SCF is a *composed* scoring rule if there is (finite) sequence of scoring rules where each scoring rules (if there is a next scoring rule) is used to break ties, (if there are any), of the previous rule.

Any scoring rule is a composed scoring rule, by taking the length of the sequence to be one.

Intuition 4.0.0.13. Roughly speaking, reinforcement characterises scoring rules (composed) — sometimes reinforcement can be used to reason about all scoring rules.

Definition 4.0.0.14 (Cancellation)**.** An SCF satisfies *cancellation* if for all A and R_N,

$$\Big(\forall x, y \in A : n_{xy} = n_{yx}\Big) \implies f(R_N, A) = A$$

NB: Cancellation is only meaningful when n is even.

Remark 4.0.0.15. Borda is a scoring rule which satisfies cancellation. To see as such, consider we can alternatively formulate the total Borda score $s_{\text{NarrowBorda}}(x, A)$ of alternative x as, the sum of the number of voters that rank $y \neq x$ below x for each alternative y, i.e.,

$$s_{\text{NarrowBorda}}(x, A) = \sum_{y \in A \setminus \{x\}} n_{xy}$$

This is just changing the order of summation from summing first over alternatives and then over voters to the reverse order. As such, we see that if for all $x, y \in A$, $n_{xy} = n_{yx}$, then clearly $s_{\text{NarrowBorda}}(x, A) = s_{\text{NarrowBorda}}(y, A)$ for all $x, y \in A$, and so $f(R_N, A) = A$.

Remark 4.0.0.16. Plurality does not satisfy cancellation. Consider the profile on three alternatives with two voters where one voter has preferences $a \succ_1 b \succ_1 c$ and the other has the reverse $c \succ_2 b \succ_2 a$. Then, for the total feasible set $A = \{a, b, c\}$, $n_{ab} = n_{ba} = 1$, $n_{ac} = n_{ca} = 1$, and $n_{bc} = n_{cb} = 1$, so the cancellation condition is satisfied, but $f(R_N, A) = \{a, c\} \neq A$.

Theorem 4.0.0.17 (Young, 1974)**.** *Borda's (narrow) rule is the only SCF satisfying neutrality, Pareto-optimality, reinforcement, and cancellation.*

[2] Meaning it is in **P**.

§4.1 Family of Condorcet Extensions

Definition 4.1.0.1. An SCF f is a *Condorcet extension* if it picks a the (strict) Condorcet winner whenever there is one, and otherwise does something (anything) else.

Can you think of a SCF which is a Condorcet extension?

4.1.1 Fishburn's Classification

Idea 4.1.1.1. Fishburn's Classification is used to categorise Condorcet Extensions based on the amount of *information* used to compute them.

Definition 4.1.1.2 (C1 Graph)**.** The *C1 Graph* or *Majority Graph* with respect to a preference profile R_N is the directed graph with vertex set U and and edge from a to b if $a \succ_M b$. That is, the edges are given by the majority relation, where there is an edge from a to b iff a strictly wins the pairwise majority comparison with b.

Definition 4.1.1.3 (C2 Graph)**.** The *C2 Graph* or *Weighted Majority Graph* is the directed graph with vertex set U and edges from a to b with weight n_{ab}.

Remark 4.1.1.4. It is clear that we can extract the C1 Graph from the C2 Graph by just looking at the edge weights and seeing whether an edges has weight greater than the reverse edge.

Definition 4.1.1.5 (Margin Graph)**.** The *Margin Graph* is the directed graph with vertex set U and an edge from a to b with weight $n_{ab} - n_{ba}$ whenever $n_{ab} > n_{ba}$.

Remark 4.1.1.6. It is clear that the margin graph has precisely the same edges as the C1 graph, but with weights given by the margin of victory.

Definition 4.1.1.7 (Fishburn's Classification)**.** Fishburn classifies Condorcet extensions (and more generally SCFs) into three classes C1, C2, and C3.

- **C1:** f depends only on the majority relation, that is it is a function of the C1 Graph.
- **C2:** f depends only on the weighted majority relation and is not in C1, that is it is a function of the C2 Graph (but not just the C1 Graph).
- **C3:** f is not a function of the C2 Graph (and hence it is not C1 nor C2).

Example 4.1.1.8 (Copeland's Rule)**.** We define Copeland's Rule as,

$$f_{\text{Copeland}}(R_N, A) := \text{argmax}_{x \in A} \left|\{y \in A : x \succ_M y\}\right|$$

That is, the Copeland rule gives those alternatives that win the most pairwise majority comparisons.

The Copeland rule is a C1 rule, since it depends only on the majority relation. We can see that the Copeland rule can be thought of as returning all those alternatives that have the maximal outgoing edges in the C1 graph.

We also note, that the Copeland Rule is a Condorcet extension, since if there a Condorcet winner, i.e. an alternative that wins all pairwise majority comparisons, then it is the unique alternative with the most outgoing edges in the C1 graph, and so it is the unique winner under the Copeland Rule.

Example 4.1.1.9 (Maximin Rule)**.** We define the Maximin rule as,

$$f_{\text{Maximin}}(R_N, A) := \text{argmax}_{x \in A} \min_{y \in A \setminus \{x\}} n_{xy}$$

That is, the Maximin rule gives those alternatives whose weakest pairwise majority victory is as strong as possible.

The Maximin rule is a C2 rule, since it depends on the weighted majority relation, but not just the majority relation. We can see that the Maximin rule can be thought of as returning all those alternatives that have the maximal weight on their weakest outgoing edge in the C2 graph.

Let us pause to think how we would formally show that it is a C2 rule. We would exhibit profiles that have identical C1 Graphs, but for which the maximin rule differs.

We also note, that the Maximin Rule is a Condorcet extension, since it must win in the majority relation, so if there is a Condorcet winner a, it must be that $n_{ay} > n_{ya}$ for all $y \neq a$, so $n_{ay} > n/2$. For any other alternative $x \neq a$, then since $n_{ax} > n/2$ and there are n voters, it must be that $n_{xa} < n/2$.

Example 4.1.1.10 (Borda's (narrow) Rule)**.** We note that, Borda's (narrow) rule is given by,

$$f_{\text{Borda}}(R_N, A) := \text{argmax}_{x \in A} \sum_{y \in A \setminus \{x\}} n_{xy}$$

Thus clearly it is a C2 rule, since it depends on the weighted majority relation. On the C2 graph, we compute the Borda winners as those alternatives that have the maximal total outgoing edge weight.

It should be noted that Borda's rule is **not** a Condorcet extension.

Think why?

Example 4.1.1.11. Plurality is a C3 rule, since it is not a function of the C2 graph.

Come up with two profiles with identical C2 Graphs but differing Plurality winners

Aside from Borda's rule what other (if any) scoring rules are in C1, C2, and C3

.

Example 4.1.1.12 (Young's Rule)**.** Returns all alternatives that can be made a (strict) Condorcet Winner when we remove a minimal number of voters. That is, over each alternative, what is the fewest number of voters we need to remove to make that alternative a Condorcet winner, and then we return those alternatives for which this number is minimal.

Young's rule is obviously a Condorcet extension. Young's rule is intractible. Young's rule is a C3 rule, for some intuition for why it is not C2, to know what voters to delete, we need to have access to the rankings in the profile (try to prove it formally).

Remark 4.1.1.13. When there are only two alternatives, majority rule (which is just plurality) is the the only non-trivial, monotonic scoring rule, and the only Condorcet extension.

Proposition 4.1.1.14 (Condorcet, 1785)**.** *Borda's rule is not a Condorcet extension when there are three or more alternatives.*

Proof. First for the case of three alternatives consider the profile given by four voters having preferences $a \succ b \succ c$, and three voters having $b \succ c \succ a$. Then a gets a total Borda score of $4 \cdot 2 + 3 \cdot 0 = 8$, and b gets a total Borda score of $4 \cdot 1 + 3 \cdot 2 = 10$. So Borda's rule does not select a. Despite this, a is the Condorcet winner, since a beats b and c by 4 votes to 3.

Now in general suppose there are $m \geq 3$ alternatives and consider the profile given by n_1 voters having preference relations $P_1 : a \succ b \succ \cdots$, and n_2 voters having preference relations $P_2 : b \succ \cdots \succ a$. For a to be a Condorcet winner, we need $n_1 > n_2$. Consider that a gets a total Borda score of $n_1 \cdot (m-1) + n_2 \cdot 0 = n_1(m-1)$, and b gets a total Borda score of $n_1 \cdot (m-2) + n_2 \cdot (m-1)$. We can thus ensure that b has a higher Borda score than a and that a is the Condorcet winner by choosing n_1 and n_2 such that $n_2 < n_1 < (m-1)n_2$, for $m \geq 3$ this can be achieved easily.

Hence, we can construct a profile where a is the Condorcet winner but b has a higher Borda score, and so Borda's rule does not select the Condorcet winner for each $m \geq 3$ (observe the inequality fails when $m = 2$). The above example for $m = 3$ does exactly this. □

Theorem 4.1.1.15. *No scoring rule is a Condorcet extension when there are three or more alternatives.*

Proof. Exercise. (Hint: Condition first on monotonic and non-monotonic scoring rules, maybe use Proposition 4.0.0.6 and then construct profiles.) □

Remark 4.1.1.16. We could think of Borda scoring as giving points per rank, and describe Borda's rule as choosing the alternatives with the highest average rank.

Theorem 4.1.1.17 (Smith, 1973)**.** *A Condorcet winner is never the alternative with the lowest Borda score. Borda's rule is the only scoring rule for which this holds.*

We can think of this as saying that Borda is the scoring rule which is most compatible with the Condorcet criterion.

Equivalently, a Condorcet loser [3] *is never the alternative with the highest Borda score (and again the only scoring rule for which this works is Borda's).*

Proof. Exercise. □

Theorem 4.1.1.18 (Gehrlein et al., 1978)**.** *When all preferences profiles are equally likely, over all scoring rules, Borda's rule maximizes the probability of selecting the Condorcet winner (when one exists).*

Idea 4.1.1.19 (Unifying Borda and Condorcet)**.** Despite there being conflict between Borda and Condorcet there have been some attempts to unify the rules.

- Black's Rule: Return the Condorcet winner when one exist, otherwise the Borda winner(s) (by definition a Condorcet extension) - it is slightly ad hoc.
- Baldwin's Rule: Run-off method based on (narrow) Borda's rule. In each round, all the minimal (narrow) Borda score alternatives are deleted, until no more deletions are possible — output this set now. Baldwin's rule is a Condorcet extension (Exercise), but it is not monotonic [4].

Theorem 4.1.1.20 (Young and Levenglick, 1978)**.** *For three or more alternative, no Condorcet extension satisfies reinforcement*

Proof. Assume for contradiction that f is an SCF which is a Condorcet extension satisfying reinforcement. Consider the profile R_N given below,

$a \succ b \succ c$	$b \succ c \succ a$	$c \succ a \succ b$
2 voters	2 voters	2 voters

[3]An alternative x is a Condorcet loser if it strictly loses all non-reflexive pairwise majority comparisons.
[4]Every scoring run-off rule violates monotonicity (Smith, 1973)

Since $f(R_N, \{a, b, c\})$ must select some alternative and the profile is symmettric in a, b, and c, there is no Condorcet winner, so without loss of generality assume $b \in f(R_N, \{a, b, c\})$.

Consider the profile $R_{N'}$ below,

$b \succ a \succ c$	$a \succ b \succ c$
2 voters	1 voter

Then, in $R_{N'}$, b is the Condorcet winner, so $f(R_{N'}, \{a, b, c\}) = \{b\}$. By reinforcement, $f(R_N \cup R_{N'}, \{a, b, c\}) = f(R_N, \{a, b, c\}) \cap f(R_{N'}, \{a, b, c\}) = \{b\}$.

Now, consider the combined profile, here a is the Condorcet winner, so $f(R_N \cup R_{N'}, \{a, b, c\}) = \{a\}$. Contradiction.

It is convenient in this proof to draw the margin graphs for R_N and $R_{N'}$, then we can easily combine them to get the margin graph for $R_N \cup R_{N'}$. □

Idea 4.1.1.21. To prove this for a general number of alternatives, we can *cheat* by adding extra preferences at the bottom of the profiles R_N and $R_{N'}$, and taking the same feasible set. Exercise: can we show that the theorem holds for the total feasible set on any number of alternatives greater than three.

Usually, when something goes wrong for some number of alternatives, it does for larger numbers of alternatives too — so it is interesting when it first goes wrong, which is usually for three alternatives (majority is axiomatically nice for two alternatives).

Idea 4.1.1.22 (Social Preferences)**.** SPF : $\mathcal{L}(U)^N \to \mathcal{F}(\mathcal{L}(U))$, where $\mathcal{F}(\mathcal{L}(U))$ is the set of non-empty subsets of $\mathcal{L}(U)$.

In this context, Young and Levenglick's theorem (Theorem 4.1.1.20) doesn't hold.

§4.2 Kemeny's Rule

Definition 4.2.0.1 (Social Preference Functions)**.** A *social preference function* (SPF) is a function h that maps a preference profile to a non-empty set of linear orders over U. That is,

$$h : \mathcal{L}(U)^N \to \mathcal{F}(\mathcal{L}(U))$$

We can think of a social preference function as a generalisation of a social welfare function, where we can output a set of orders instead of just one, and where the orders must be linear orders instead of just weak orders.

Definition 4.2.0.2 (Kemeny's Rule)**.** Kemeny's rule is a SPF defined as,

$$h_{\text{Kemeny}}(R_N) := \text{argmax}_{\succsim \in \mathcal{L}(U)} \sum_{i \in N} |\{\langle x, y \rangle : x \succ y, x \succ_i y\}|$$

As with SCFs, we approach them from an optimization perspective. Here, instead of optimizing over alternatives, we optimize over linear orders.

Informally, for each linear order, we give it points based on how much it agrees with a voters preference, and sum over all voters, and then we return (all) those linear orders that have the highest total score.

NB: We can think about Kemeny's rule as that which yields all rankings that maximize agreement with the profile. More intuitively, we can think about it minimizing the pairwise disagreement as measured by the bubble-sort distance.

Example 4.2.0.3. For some linear order say $a \succ b \succ c$, we can compute the Kemeny-Score as,

$$\text{Kemeny-Score}(\begin{pmatrix} a \\ b \\ c \end{pmatrix}) = n_{ab} + n_{ac} + n_{bc}$$

It is clear that there is an exponential number of linear orders, thus an exponential number of Kemeny-Scores we must compute, so it is not the easiest to compute Kemeny's rule by brute-force.

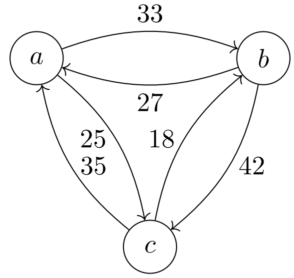
Also, note that in some sense (despite it being a SPF) Kemeny's rule only relies on the C2 graph, since we can compute the Kemeny-Score of a linear order just by looking at the edge weights in the C2 graph.

We can think of Kemeny's rule as picking the maximal weight acyclic selection of direction on all pairs of the edges [5]. What if we didn't care for acyclicity? We would then always pick the direction of the edge with the higher weight, and in some sense this gives us an *ideal* score, upper bounding the Kemeny-Score — that is, if cyclic rankings were *allowed*, $\gtrsim_M$ would maximize the Kemeny-Score.

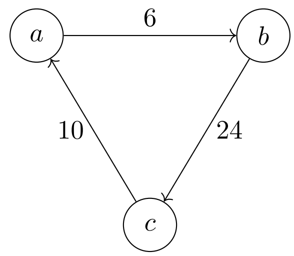
We can think about starting with the majority cycle (described above) and then massaging this into a linear order by changing the direction of some edges, any change invokes a penalty in the Kemeny-Score of $n_{xy} - n_{yx}$, precisely the edge weight in the margin graph.

Example 4.2.0.4. Consider the following profile, given with its C2 graph below, and its margin graph below that.

23	17	10	8	2
a	b	c	c	b
b	c	a	b	a
c	a	b	a	c



[5] An acyclic tournament is a linear order.



We look at the C2 graph, and notice that the *ideal* majority score (when we disregard the acyclicity condition) is given by $33 + 42 + 35 = 110$.

We can consider the linear order $a \succ b \succ c$, and observe (via the margin graph) that this incurs a penalty of 10 for the edge $a \succ c$. It is not too hard to see, that this means that we get the Kemeny-Score of $110 - 10 = 100$ for this linear order. Perhaps, it is not too much of a stretch to notice that this is not the highest Kemeny-Score we can get, since we can instead incur the 6 penalty for the edge $a \succ b$ by considering the linear order $b \succ c \succ a$.

Remark 4.2.0.5. We can then reduce the problem of computing the Kemeny winners, to a problem in the margin graph. Namely, we find sets of edges that are minimal in total weight, and such that the graph produced by reversing these edges (and keeping the rest of the margin graph as is) is acyclic.

Idea 4.2.0.6. This idea of picking a subset of edges, reversing the direction of that subset, and then checking acyclicity of the resulting graph, is a little bit of a pain to think about. Instead, we would like to think about deleting these edges instead, and the following lemma helps with this.

Lemma 4.2.0.7. *Let $G = (V, E)$ be a directed graph, and $E' \subseteq E$ be a subset of edges. Then, G can be made acyclic by inverting a subset $E'' \subseteq E'$ iff $(V, E \setminus E')$ is acyclic.*

Proof.
$\implies$: Easy, remove all of E'.

$\impliedby$: Greedily insert edges in E' back in, one after another. At each step orient the edges so that no cycle forms. If at any stage, both orientations yield a cycle, then the previous graph was cyclic. Contradiction. ☐

Comment. This formulation is not as clean as we may hope, but is necessary, for example, consider the three cycle graph, deleting the whole cycle makes is acyclic, but reversing it does not.

Remark 4.2.0.8. This lemma, allows us to consider the problem now to be, find a subset of edges E' with minimal total weight, such that $(V, E \setminus E')$ is acyclic.

We might hope this to now be rather easy, "For each cycle, remove the minimum weight edge.", but cycles may overlap, and so this is slightly more tricky.

Comment. Recall, we showed that reinforcement is satisfied by all scoring rules, and no Condorcet extensions — redefining these properties for SPFs, we can show that there is a Condorcet extension that satisfies reinforcement, namely, Kemeny's rule (cheating, of course by redefining the properties in this context).

Definition 4.2.0.9 (Young's Characterization)**.**
An SPF h satisfies *reinforcement* if for all disjoint sets of voters N and N', and all profiles $R_N \in \mathcal{L}(U)^N$ and $R_{N'} \in \mathcal{L}(U)^{N'}$, where $h(R_N) \cap h(R_{N'}) \neq \emptyset$, then $h(R_N \cup R_{N'}) = h(R_N) \cap h(R_{N'})$ [6].

Two alternatives x and y are *adjacent* in a preference relation $\succsim$ if there is no alternative z such that $x \succ z \succ y$ or $y \succ z \succ x$. We say an SPF h satisfies *Condorcet-consistency* if for all profiles R_N, $\succsim \in h(R_N)$, and $x \succ y$

[6] Again, formally, we are considering this notion of an expanded domain.

where x and y are adjacent in $\gtrsim$, we have $x \succ_M y$. Moreover, $x \sim_M y$ implies $(\gtrsim \setminus \{\langle x, y\rangle\}) \cup \{\langle y, x\rangle\} \in h(R_N)$ [7].

Remark 4.2.0.10. Consider that Condorcet-consistency, implies the natural notion that we might think of as a Condorcet extension in the context of SPFs. Namely, if there is a Condorcet winner and an SPF is Condorcet-consistent, then the Condorcet winner is ranked above all other alternatives in every linear order given by the SPF.

Why?

Theorem 4.2.0.11 (Young and Levenglick, 1978)**.** *Kemeny's Rule is the only neutral SPF satisfying reinforcement and Condorcet-consistency.*

Remark 4.2.0.12. Without going into the proof of the *only* part of the above theorem, it is a good exercise to think why Kemeny's rule is neutral and satisfies reinforcement and Condorcet-consistency.

Comment. Often, we want to reason in the course about profiles by way of their C1, C2, or margin graphs. It is often easier to go straight from a graph, and we may not first want to construct a profile which yields that graph.

Theorem 4.2.0.13 (McGarvey's Theorem)**.** *For tournament $G = (V, E)$, there exists a preference profile R_N such that the C1 graph with respect to R_N is G. Further, the same profile yields a margin graph (the same graph) with edge weights of* 1*.*

Proof. See lectures for proof. □

Definition 4.2.0.14 (Feedback Arc Set)**.** The problem *Feedback Arc Set* (FAS) is the problem: is it possible to make a directed graph acyclic by removing k edges.

Theorem 4.2.0.15. *FAS is NP-Complete (Karp, 1972). FAS is NP-Complete when restricted to tournament graphs (Alon, 2006; Charbit et al. 2007; Conitzer 2006), refer to this problem as* ***T-FAS****.*

Theorem 4.2.0.16 (Bartholdi et al., 1989)**.** *Deciding whether a there exists a ranking with Kemeny score at least s is NP-Complete.*

Proof. Slightly tricky reduction from T-FAS using McGarvey's theorem and Lemma 4.2.0.7. Exercise. □

Remark 4.2.0.17. The Kemeny score problem is at least as hard as T-FAS. Finding a Kemeny ranking is at least as hard as the Kemeny score problem, so finding a Kemeny ranking is NP-Hard. Finding a Kemeny ranking is NP-Hard in the restricted case for even $n \geq 4$ and odd $n \geq 7$. If it is NP-Hard to find a Kemeny ranking for $n = 3, 5$ is open. Deciding if a given ranking is a Kemeny ranking is CoNP-Complete (clearly). There are approximation algorithms for finding a Kemeny ranking.

[7] This final condition on $x \sim_M y$ is more of a technicality which says that if there is a tie and one of the linear orders given by h has $x \succ y$, then we can flip the order of x and y in that linear order and it will still be given by h.

5

Weighted Tournament Solutions

Comment. In this chapter, we consider C2 social choice functions.

Notation 5.0.0.1. Given a preference profile R_N, and alternatives $x, y \in U$, let $m(x, y) := n_{xy} - n_{yx}$ be the majority margin between x and y. Clearly $m(x, y) = -m(y, x)$, and $m(x, x) = 0$.

Definition 5.0.0.2 (Weighted Tournament Solution)**.** An SCF is a *weighted tournament solution* if it only depends on $\big(m(x, y)\big)_{x,y \in U}$. Often, all we need is the order among $\big(m(x, y)\big)_{x,y \in U}$ (note, this is not the case for Kemeny). Basically, this is the same as a function of the margin graph.

Theorem 5.0.0.3 (Debord's Theorem)**.** *The following generalises McGarvey's Theorem.*

Let $\big(m(x, y)\big)_{x,y \in U}$ *be a collection of integers with* $m(x, y) = -m(y, x)$ *for all* $x, y \in U$*, and* $m(x, x) = 0$ *for all* $x \in U$*. Then, there exists a preference profile* R_N *with majority margins* $\big(m(x, y)\big)_{x,y \in U}$ *if and only if for all* $x \neq y$*,* $m(x, y)$ *has the same parity.*

The parity condition makes sense, consider first that $n_{xy} + n_{yx} = n$ *and so* $m(x, y) = n_{xy} - n_{yx} = 2n_{xy} - n$*, which has the same parity as* n*.*

Proof. Exercise. □

§5.1 Clones

Example 5.1.0.1. In 1969, Thunder Bay, Ontario, Canada, held a plurality referendum to decide its name.

- Thunder Bay: 15,870 votes
- Lakehead: 15,302 votes
- The Lakehead: 8,377 votes

This phenomena is vote splitting, or manipulation by cloning.

Plurality is especially, susceptible to this, something like instant run-off would not fall to this so easily.

Definition 5.1.0.2 (Clones)**.** For a preference profile R_N a subset $C \subseteq U$, s.t. $|U| > |C| \geq 2$ is a set of *clones* if for all $c, c' \in C$, $x \in U \setminus C$, and $i \in N$ it holds that $c \succ_i x$ iff $c' \succ_i x$.

Informally, a set of clones are always ranked next to each other by every voter, though the ranking among the clones may be arbitrary (clones appear as a contiguous interval in each voters preference relation).

Observation 5.1.0.3 (Tideman, 1987). Most SCFs are susceptible to manipulation by adding or deleting clones.

Definition 5.1.0.4 (Independence of Clones)**.** An SCF f satisfies *independence of clones* (IoC) if for all profiles R_N, feasible set $A \in \mathcal{F}(U)$, and set of clones $C \subseteq A$, and $c \in C$ the following two properties hold.

1. For all $a \in A \setminus C$: $a \in f(R_N, A) \iff a \in f(R_N, A \setminus \{c\})$.
2. $C \cap f(R_N, A) \neq \emptyset$ if and only if $C \cap f(R_N, A \setminus \{c\}) \neq \emptyset$.

The first property says that adding or removing a clone alternative should not affect the whether a non-clone is selected or not.

The second says that adding or removing a clone from a clone group does not affect whether or not the clone group contains a winner (note we specify clone groups of size at least 2).

Example 5.1.0.5. Instant-runoff satisfies independence of clones, but we are not too fond of it because it fails to satisfy many other desirable properties: it is not monotonic, nor a *Condorcet extension.*

§5.2 Ranked Pairs

Definition 5.2.0.1 (Ranked Pairs: Tideman, 1987)**.** The SCF *Ranked Pairs* was proposed to achieve independence of clones. It may be viewed as a greedy heuristic for finding rankings with a high Kemeny score.

Ranked Pairs is an SCF that follows more algorithmically.

1. Order the edges of the majority graph by decreasing weight, use a tiebreaker if necessary (often we assume that their are no ties and also there are no majority ties so that there is an edge between every pair in the majority graph). Restrict to just the edges with both endpoints in the feasible set A.
2. Initialize an empty graph G on the vertex set of all feasible alternatives A.
3. Process the edges as follows in the order given by (1). If $\langle a, b\rangle$ is the next majority edge to be processed:
   - If $G \cup \{\langle a, b\rangle\}$ is acyclic, add $\langle a, b\rangle$ to G.
   - Otherwise, skip $\langle a, b\rangle$ (Alternatively, add $\langle b, a\rangle$ to G).

This gives a linear order on the feasible alternatives. Finally, return the highest ranked alternative in this order, *the ranked pairs winner.*

Clearly, this is polynomial-time. Essentially we are greedily adding edges where we can starting with the ones that give the highest greedy Kemeny score (may not be optimal).

Why does this algorithm always output something?

Theorem 5.2.0.2 (Tideman and Zavist, 1989)**.** *When equipped with a suitable tiebreaker, Ranked Pair satisfies IoC* [1]*.*

Given a tiebreaker, Ranked Pairs is a resolute SCF and can be computed in polynomial-time.

[1]Caveat: The tiebreaker violates neutrality and/or anonymity

Definition 5.2.0.3 (PUT Ranked Pairs)**.** Parallel-Universes Tiebreaking. The irresolute SCF that outputs all alternatives that can win under some tiebreaker, we then can maintain anonymity and neutrality (but annoyingly, does not satisfy independence of clones). Computing it is NP-Hard (reduction from SAT).

Definition 5.2.0.4 (Stability for Winners: Holliday and Pacuit, 2023)**.** An SCF satisfies *stability for winners* if for all profiles R_N, and all feasible sets A and $b \in U \setminus A$: if $a \in f(R_N, A)$ and $a \succ_M b$, then $a \in f(R_N, A \cup \{b\})$.

Remark 5.2.0.5. Stability for winners implies that there can not be a *spoiler effect* as in the Gore, Bush, Nader case (2000 US Presidential Election).

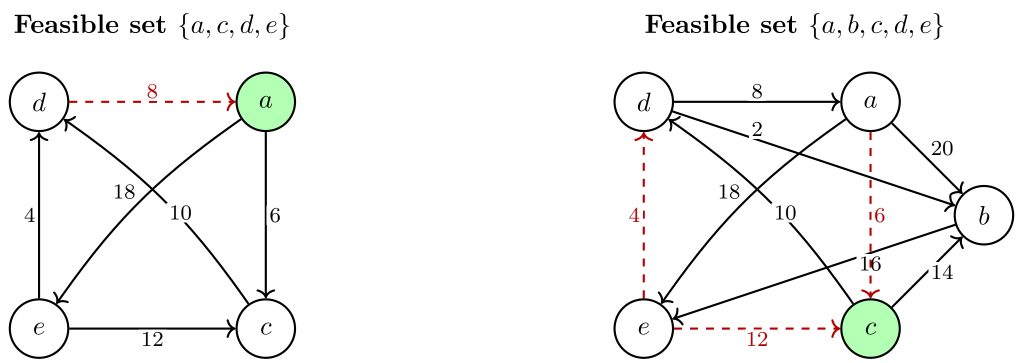


Figure 5.1: Ranked Pairs fails stability for winners. Solid arrows are the majority edges that are included in the ranked pairs algorithm; dashed red arrows are rejected edges (would create a cycle). Winner shown in green. Removing non-winner b changes the winner from c to a.

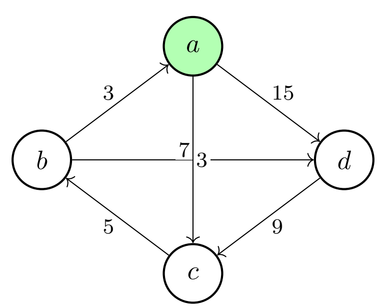
Example 5.2.0.6 (Ranked Pairs Fails Stability for Winners)**.**

Definition 5.2.0.7 (The (strong) no show paradox)**.** Consider the following situation. A profile R_N has that x is a winner under f. A voter $i \notin N$ ranks x first. The new profile $R_{N \cup \{i\}}$ no longer gives that x is the winner.

It is a different type of manipulability, where it would be beneficial for a voter to not participate in an election.

This is known as the *(strong) no show paradox*.

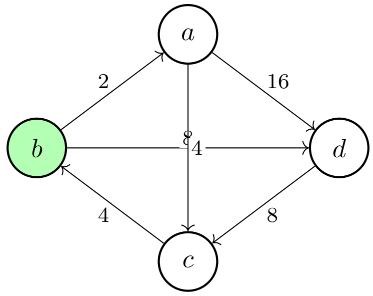
Example 5.2.0.8 (Ranked pairs is susceptible to the no show paradox)**.** Consider the following margin graph.



The ranked pairs algorithm will select all but the two 3 edges, and give that a is the winner.

Consider the profile now with a new voter i with preference relation $a \succ_i b \succ_i c \succ_i d$.

The new profile looks like,



And we have winner, b.

Definition 5.2.0.9 (Positive Involvement)**.** An SCF f satisfies positive involvement if for all feasible sets A and alternatives $x \in A$, and all profiles R_N and $R_{N'}$ where $N' = N \cup \{i\}$, where $\text{Max}(\succsim_i, A) = x$: if $x \in f(R_N, A)$, then $x \in f(R_{N'}, A)$.

§5.3 Split Cycle

Definition 5.3.0.1. Consider a preference profile and let $C = (x_1, \ldots, x_k)$ be a simple cycle in the margin graph [2], when restricted to a feasible set A. The splitting number $\text{sp}(C)$ is the smallest margin between consecutive alternatives in C.

We say that a *defeats* b if $a \succ_M b$ and for every (simple) cycle C containing a and b, $\text{sp}(C) < m(a, b)$.

Informally, we think that a defeats b if it strictly wins in the majority relation, and further the edge from $a \to b$ is never the weakest link on a cycle.

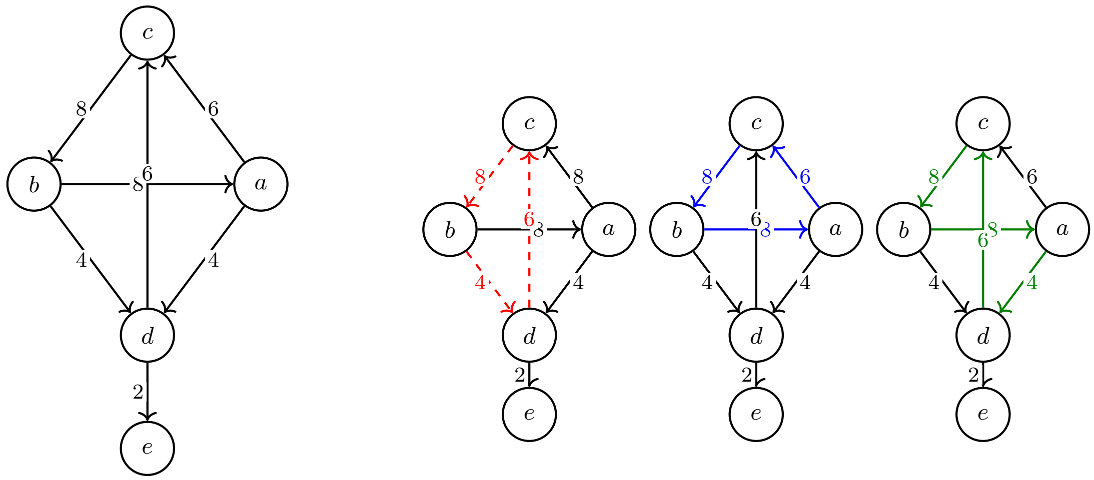
Definition 5.3.0.2 (Split Cycle)**.** The social choice function *Split Cycle* is the function that returns all undefeated candidates.

Denote it by f_{SC}.

We can think of the computation of split cycle as simply starting with the margin graph, identifying every simple cycle in the margin graph and deleting the lightest edge along each cycle, finally it outputs those alternatives with no incoming edges exactly.

Example 5.3.0.3. Consider the following margin graph, in colours each simple cycle is highlighted. The Split Cycle algorithm would remove those edges corresponding to the lightest weight in the cycle, and then output the candidates that have no incoming edges — these are precisely the undefeated candidates. The remaining edges give exactly the defeat relation.

[2]A path as opposed to a walk.



Red: cycle (c, b, d), $\text{sp} = 4$ Blue: cycle (c, b, a), $\text{sp} = 6$ Green: cycle (c, b, a, d), $\text{sp} = 4$

It is clear, that the defeat relation is **not** transitive, since after deleting these three lightest edges, c defeats b and b defeats a, but c does not defeat a.

Lemma 5.3.0.4 (Defeat Relation is acyclic)**.** *We can think of* defeat *as a relation on alternatives. To show that Split Cycle gives a non-empty choice set, it suffices (but is not necessary) to show that the defeat relation is acyclic.*

Proof. Suppose for contradiction that there is a cycle under the defeat relation, then there is a simple cycle under the the defeat relation, say $a_1, \ldots, a_k$. Then, by definition of the defeat relation, $C := a_1, \ldots, a_k$ is a simple cycle in the margin graph. Then, consider that for each edge $a_i \to a_{i+1}$ (incl. the edge $a_k \to a_1$), in the defeat cycle, by definition of defeat, we have that $m(a_i, a_{i+1}) > \text{sp}(C)$. Consider that by definition of splitting number, there is some edge $a_j \to a_{j+1}$ in C such that $m(a_j, a_{j+1}) = \text{sp}(C)$, contradicting the fact that $m(a_j, a_{j+1}) > \text{sp}(C)$. $\square$

Observation 5.3.0.5. To check whether a defeats b, we only need to check the simple cycles containing the edge $a \to b$. That is we need not check cycles containing a and b when they are not adjacent, and certainly not all simple cycles in the margin graph. We omit proof.

Definition 5.3.0.6. For alternatives $a, b \in U$ with $a \succ_M b$, define the *cycle number* to be

$$\text{cyc}(a, b) := \max \Bigg(\{0\} \cup \{\text{sp}(C) : \text{C a simple cycle in margin graph containing edge}\langle a, b\rangle\} \Bigg)$$

Informally, the cycle number of a and b is the the maximum splitting number among simple cycles containing the edge $a \to b$, or 0 if there are no such cycles.

Lemma 5.3.0.7. *a defeats b if and only if $m(a, b) > \text{cyc}(a, b)$.*

Proof. Exercise. $\square$

Remark 5.3.0.8. The naive approach we give to compute f_{SC} requires checking all simple cycles in the margin graph, which is exponential in the number of alternatives. We hope to use the above ideas, to get a more computationally appealing method to compute f_{SC}.

Definition 5.3.0.9. The *strength* of a path in the margin graph is the smallest margin between consecutive alternatives on the path.

Let $\text{strength}(x, y)$ denote the strength of the strongest path from x to y.

Lemma 5.3.0.10. *a defeats b if and only if $a \succ_M b$ and* $\text{strength}(b, a) < m(a, b)$*.*

Remark 5.3.0.11. We characterise the notion of defeat via strength because $\text{strength}(\cdot, \cdot)$ is a notion that *can* be computed efficiently using a variant of the Floyd-Warshall algorithm ($O(|A|^3)$). Thus, we can compute split cycle winners in polynomial time. Explicitly we do this by,

1. Compute the margin graph from the profile: $O(|A|^2 n)$
2. Compute for each pair of alternatives $\text{strength}(a, b)$: $O(|A|^3)$ each so $O(|A|^5)$ total.
3. For each pair, check if a defeats b by checking if $a \succ_M b$ and $\text{strength}(b, a) < m(a, b)$: $O(|A|^2)$.
4. Return all those alternatives with no incoming edges in the defeat graph.

Proposition 5.3.0.12 (Split Cycle satisfies stability for winners)**.**

Proof. Suppose for contradiction that split cycle fails stability for winners, then there is a profile R_N, feasible set A, and $b \in U \setminus A$ such that there is some $a \in f_{\text{SC}}(R_N, A)$ and $a \succ_M b$, but $a \notin f_{\text{SC}}(R_N, A \cup \{b\})$.

Consider that adding b to the set of feasible alternatives can only add paths in the margin graph between vertices, so $\text{strength}_{A\cup\{b\}}(x, y) \geq \text{strength}_A(x, y)$ for all $x, y \in A$. For contradiction, suppose a is defeated in $A \cup \{b\}$, then there is some alternative c such that $\text{strength}_{A\cup\{b\}}(a, c) < m(c, a)$. But $\text{strength}_{A\cup\{b\}}(a, b) \geq \text{strength}_A(a, b)$, thus $\text{strength}_A(a, c) < m(c, a)$. If $c \in A$, then a is defeated in A, contradicting that $a \in f_{\text{SC}}(R_N, A)$. Thus, $c = b$, but this gives a cyclic strict majority between a and b, since $a \succ_M b$, giving contradiction. $\square$

Proposition 5.3.0.13 (Split Cycle satisfies positive involvement)**.**

Proof. Exercise. $\square$

Proposition 5.3.0.14 (Split Cycle satisfies independence of clones)**.**

Proof. Omitted. $\square$

Proposition 5.3.0.15 (Split Cycle subsumes ranked pairs)**.** *If f_{RP} is the ranked pairs SCF, then for all profiles R_N and feasible sets A, $f_{RP}(R_N, A) \subseteq f_{SC}(R_N, A)$.*

Sketch. We omit the details of tie-breaking.

Suppose that $a \notin f_{\text{SC}}(R_N, A)$, then there is some b that defeats a. Then, Lemma 5.3.0.10 gives that $b \succ_M a$ and $\text{strength}(b, a) < m(a, b)$. Now consider the cycle formed by the traversal from $b \to_M a$ and then along the strongest path from a to b. At some point the ranked pairs algorithm will encounter the edge $b \to a$ and will add it because, all cycles containing $b \to a$ have splitting number less than $m(a, b)$, so the ranked pairs algorithm will never reject the edge $b \to a$, and thus a can not be the winner under ranked pairs. $\square$

Comment. To understand the trade-offs between Ranked Pairs and Split Cycle, we briefly look at the following impossibility results.

First, note that both Ranked Pairs and Split Cycle satisfy *ordinal margin invariance*; that is they depend only on the ordering of the margins $(m(x, y))_{x,y\in U}$, not on the actual values of the margins. Note, Kemeny does care about the numbers not just their ordering.

Theorem 5.3.0.16 (Holliday, 2023)**.** *The following are incompatible: Condorcet extension, Condorcet loser criterion, ordinal margin invariance, positive involvement, and resolvability.*

Where by Condorcet loser criterion *we mean that we never select a Condorcet loser. And, by* resolvability *we mean that for all profiles R_N and feasible sets A, when $|f(R_N, A)| > 1$, there is some profile $R_{N \cup \{i\}}$ such that $|f(R_{N \cup \{i\}}, A)| = 1$ (we can always add one voter to break a tie).*

Remark 5.3.0.17. Split Cycle satisfies all but resolvability, while Ranked Pairs satisfies all but positive involvement.

Definition 5.3.0.18 (Quasi-Resoluteness)**.** An SCF f is *quasi-resolute* if for all profiles R_N and feasible sets A, where the edges of the margin graph are all distinct, then $|f(R_N, A)| = 1$.

Such profiles are sometimes called *uniquely weighted*. It is clear that they do not permit majority ties, i.e. for all $x \neq y$, $m(x, y) \neq 0$.

Theorem 5.3.0.19 (Holliday et al., 2024)**.** *There is no anonymous and neutral SCF that satisfies stability for winners and quasi-resoluteness.*

6

Multiwinner Voting Rules

Idea 6.0.0.1. What does multiwinner voting mean? So far, we have modelled alternatives as mutually exclusive outcomes, and we wanted to the chose the *best* alternative, or rank all alternatives. Of course, many rules we have discussed are irresolute, but this is not exactly desired behavior — our framework ushered us away from multiple winners (implicitly we imagine some out of framework tie breaker).

What if instead, we imagine we *must* elect a a specifically sized (k) set of of alternatives.

In principle, we could model multiwinner elections with the tools we have thus far, we could have voters just rank their preferences where candidates are different k-subsets of alternatives (different committees) — but this gives exponential number of alternatives, and thus is not computationally appealing, nor natural.

Another idea is to use a social welfare function, and pick the top k alternatives in the social welfare ordering, we revisit this later.

Remark 6.0.0.2. We consider two different frameworks for multiwinner voting, whereby voters *indirectly* express preferences over committees.

- The *Apportionment setting*: candidates are grouped into *parties* and voters can support a single party only. The underlying assumption is that: voters only care about how many candidates from their party are selected.

  This models, *party-list election*, which are often used in parliamentary elections. Intuitively, it seems clear that it kind of limits the expressive power of voters.

- The *Approval-Based setting*, voters can approve whichever candidates, but they do not rank them (still less expressive than our previous settings). The assumption is that voters only care about how many candidates that they approve of are selected.

In both settings, we are interested in a notion of *proportional representation.*

§6.1 Apportionment Setting

Definition 6.1.0.1. In the apportionment setting, we have a set of parties $P = \{P_1, \ldots, P_p\}$, and each voter votes for exactly one party.

For each party P_i let v_i denote the number of votes it receives. The goal is to allocate a given number of seats among the parties, *proportional* to the votes.

An instance of the apportionment problem is given by the pair $(\vec{v}, h)$ where $\vec{v} = (v_1, \ldots, v_p) \in \mathbb{N}^p$ is the *vote distribution* and $h \in \mathbb{N}_{>0}$ is the *committee size* or *house size*.

We denote the total number of votes by $v_+ := \sum_{i=1}^{p} v_i$.

The seat distribution is the vector $\vec{x} = (x_1, \ldots, x_p) \in \mathbb{N}^p$ where x_i is the number of seats allocated to party P_i, and $\sum_{i=1}^{p} x_i = h$.

Remark 6.1.0.2. An equivalent problem: For each state in a union, allocate a number of seats proportional to the population of the state.

Example 6.1.0.3. $\vec{v} = (7270, 1230, 220); h = 21$, and a seat distribution (with respect to $\vec{v}$ and h) is $\vec{x} = (15, 4, 2)$.

Definition 6.1.0.4 (Apportionment Method)**.** An *apportionment method* M maps every instance $(\vec{v}, h)$ to a non-empty set $M(\vec{v}, h)$ of seat distributions.

Definition 6.1.0.5 (Quota)**.** Ideally, we would like to give each party its exact *proportional share* of seats. For $(\vec{v}, h)$, the *quota* of party P_i is given by

$$q_i := \frac{v_i}{v_+} \cdot h$$

Of course, this is not in general an integer.

We call the fraction $\frac{v_+}{h}$ the *Hare quota*, and see that equivalently $q_i = \frac{v_i}{\text{Hare quota}}$.

Definition 6.1.0.6 (Hamilton Method)**.** The *Hamilton method* has two steps.

1. Calculate quotas and give each party $\lfloor q_i \rfloor$ seats. (Note, the sum of the floor of these quotas is $\leq h$).

2. Give the remaining r seats (if any) to each of the r parties with the largest fractional part of their quota (largest $q_i - \lfloor q_i \rfloor$).

Definition 6.1.0.7 (D'Hont Method)**.** Find an integer divisor d, such that

$$\sum_{i=1}^{p} \left\lfloor \frac{v_i}{d} \right\rfloor = h$$

Then, we give party P_i $\left\lfloor \frac{v_i}{d} \right\rfloor$ seats.

Such a divisor always exists when the number of votes for each party are pairwise distinct, and while it need not exist uniquely, it always apportions the same seat distribution.

Remark 6.1.0.8. The seat distribution found by the D'Hont method can be constructed by the following method.

- Define the j-th divisor as $d(j) = j$

- Construct the table with numbers $\frac{v_i}{d(j)}$ for $i \in [p]$ and $j \in \mathbb{N}$

- Choose the largest h entries of this table and distribute seats accordingly

parties	P_1	P_2	P_3
votes (v_i)	7270	1230	2220
$v_i/1$	7270	1230	2220
$v_i/2$	3635	615	1110
$v_i/3$	2423	410	740
$v_i/4$	1818	308	555
$v_i/5$	1454	246	444

Figure 6.1: Sequential D'Hondt method table.

Why does this work? (Think about it.)

Remark 6.1.0.9. We can choose the values of the divisors $d(j)$ differently, to get other apportionment methods, known as *divisor methods*. For example, the Sainte-Laguë method uses $d(j) = 2j - 1$, which can lead to different distributions.

6.1.1 Axioms for Apportionment Methods

Definition 6.1.1.1 (Basic Apportionment Properties)**.**

- *Impartiality*: the only information that matters is the vote distribution $\vec{v}$ and the house size h.
- *Exactness*: if all the quotas are integers, then the apportionment method gives each party its quota ($x_i = q_i$).
- *Homogeneity*: if all votes are multiplied by the same positive factor, the outcomes do not change (the only thing that matters are the quotas).
- *Concordance*: a party with strictly more votes than another, cannot be allocated fewer seats than the other, i.e. $v_i > v_j \implies x_i \geq x_j$.

All the rules we have seen, satisfy these properties, in fact most natural apportionment methods satisfy these properties.

Idea 6.1.1.2. We will focus on slightly harder properties to satisfy: *quota property*, *committee monotonicity*, and *population monotonicity*.

6.1.2 Quota Property

Definition 6.1.2.1 (Quota Property)**.** An apportionment method M satisfies the *quota property* if for all instances $(\vec{v}, h)$, for all $x \in M(\vec{v}, h)$, $x_i \in \{\lfloor q_i \rfloor, \lceil q_i \rceil\}$.

Example 6.1.2.2.

- The Hamilton method satisfies the quota property (reason why?).
- The D'Hondt method does not satisfy the quota property, see the following example.

party	P_1	P_2	P_3
votes (v_i)	7270	1230	2220
quota (q_i)	14.92	2.52	4.56
seats (x_i)	16	2	4

$h = 22$

Definition 6.1.2.3 (Lower Quota)**.** The weaker axiom *lower quota* requires that for all instances $(\vec{v}, h)$, for all $x \in M(\vec{v}, h)$, $x_i \geq \lfloor q_i \rfloor$.

Theorem 6.1.2.4 (Balinski and Young, 1982)**.** *D'Hondt's method satisfies the lower quota property. Moreover, it is the only divisor method that satisfies the lower quota property.*

6.1.3 Committee Monotonicity

Example 6.1.3.1 (The Alabama Paradox)**.** Increasing the committee size may result in some party having fewer seats.

party	P_1	P_2	P_3
votes (v_i)	7270	1230	2220
quota (q_i)	14.24	2.41	4.35
seats (x_i)	**14**	**3**	**4**

$h = 21$ $\rightarrow$

party	P_1	P_2	P_3
votes (v_i)	7270	1230	2220
quota (q_i)	14.92	2.52	4.56
seats (x_i)	**15**	**2**	**5**

$h = 22$

Definition 6.1.3.2 (Committee Monotonicity)**.** An apportionment method M satisfies *committee monotonicity* if this can never happen.

Example 6.1.3.3. The Hamilton method fails committee monotonicity (see the above example), but the D'Hondt method satisfies committee monotonicity, which can be seen by the tabular definition of D'Hondt's method (and likewise for all divisor methods).

6.1.4 Population Monotonicity

Example 6.1.4.1 (The Population Paradox)**.** A party that loses votes may gain seats at the expense of a party that gains votes. Formally, there are two vote distributions $\vec{v}$ and $\vec{v}'$ such that,

- $v_i < v_i'$ and $v_j > v_j'$ for some $i \neq j$.
- For some $x \in M(\vec{v}, h)$ and $x' \in M(\vec{v}', h)$, $x_i > x_i'$ and $x_j < x_j'$.

That is, we cannot have party i gaining votes and party j losing votes resulting in party i losing seats and party j gaining seats.

Definition 6.1.4.2 (Population Monotonicity)**.** An apportionment method M satisfies *population monotonicity* if this can never happen. Note there are no requirements on the other parties, they can gain or lose votes as they please.

Example 6.1.4.3. The Hamilton method fails population monotonicity, but the D'Hondt method satisfies it.

Theorem 6.1.4.4 (Balinski and Young, 1982)**.** *There is no concordant apportionment method that satisfies both the quota property and population monotonicity.*

Remark 6.1.4.5*.* A more natural notion is called *vote monotonicity*, which requires that if a party gains votes (and the rest of the parties recieve the same number of votes), it should not lose seats.

It is not known, if there is a apportionment method that satisfies the quota property, committee monotonicity, and vote monotonicity (open problem).

§6.2 Approval-based Multiwinner Voting

Comment. We need to flesh out what is even meant by *proportional representation* in this setting.

Idea 6.2.0.1. Voters express their preferences with *approval votes.* Think of a list of the candidates and each voter puts a checkmark against the ones they approve of, and leaves the rest blank.

Definition 6.2.0.2 (Approval-Based Multiwinner Rule)**.** Suppose we have a finite set of candidates C, and a finite set of voters $N = \{1, \ldots, n\}$. We let $A_i \subseteq C$ denote the set of candidates that voter i approves of, note A_i is free to be any subset of C.

A *committee* of size k is a k-sized subset of C, most often we denote it $W \subseteq C$.

Given a *preference profile* $A_N = (A_1, \ldots, A_n)$ (note our notion of preference profile here does *not* coincide with our previous notion of preference profile), and a committee size k, an *approval-based multiwinner rule* is a function that maps (A_N, k) to a non-empty set of committees [1] of size k.

6.2.1 Approval-based Multiwinner Rules

Definition 6.2.1.1 (Approval Voting (AV))**.** The *approval voting* rule, computes the number of voters approving each candidate, then outputs the k candidates with the most approvals.

Notation 6.2.1.2. For $c \in C$, let $N_c = \{i \in N : c \in A_i\}$ denote the set of voters that approve of candidate c (*approvers*), and let $n_c = |N_c|$ denote the number of voters that approve of candidate c (*approval score*).

[1]As usual we map to a set instead of a singleton for reason of symmetry

Example 6.2.1.3. What problems arise from such a system. We can consider that there may be an alternatives ranked as having the $k+1$-th approval scores. In fact such an alternative could be very close to the k-th approval score, but it is not selected. We aim for some notion of proportional representation, but this doesn't feel particularly proportional.

For a concrete example suppose: $C = \{a, b, c, x, y\}$, $k = 3$, and $|N| = 9$.

Suppose we have a profile where 5 voters approve of a, b and c only, and 4 voters approve of x and y only. Then, a, b, and c are the winners under approval voting, but x and y are not selected, even though they have a very high approval score (4 out of 9).

The above is an example of a *dictatorship of the majority*, where the majority gets all the seats, and the minority gets none.

Definition 6.2.1.4 (Satisfaction-AV (SAV))**.** The *satisfaction-approval voting* rule, chooses committee(s) W that maximize the total satisfaction:

$$\sum_{i \in N} \mathrm{sat}(i, W)$$

Where $\mathrm{sat}(i, W) = \frac{|A_i \cap W|}{|A_i|}$.

Informally, $\mathrm{sat}(i, W)$ is the fraction of candidates that voter i approves of that are selected in committee W, and we optimise maximising the sum of this quantity over all voters.

Example 6.2.1.5. In the above example, the satisfaction-approval voting rule would select all the committees with one candidates from $\{a, b, c\}$ and one candidate from $\{x, y\}$ (which seems weird).

Remark 6.2.1.6. We can conceptualise SAV by considering each voter having weight 1 which they distribute equally to the candidates they approve of, and then we select the k candidates with the highest total weight.

Definition 6.2.1.7 (Greedy-AV)**.** Iteratively construct the committee by at each step adding a candidate that is approved by the most voters that do not yet have a member they approve of in the partially constructed committee.

We say that a voter i is represented in W if $A_i \cap W \neq \emptyset$. If all voters are represented, we can add any candidates we like to fill up the committee.

Example 6.2.1.8. In the above example the Greedy-AV rule would select all the committees with one candidates from $\{a, b, c\}$ and one candidate from $\{x, y\}$.

Definition 6.2.1.9 (Proportional Approval Voting(PAV))**.** Choose committee(s) W maximizing $\mathrm{score}(W) = \sum_{i \in N} \mathrm{score}(i, W)$, where,

$$\mathrm{score}(i, W) = 1 + \frac{1}{2} + \cdots + \frac{1}{|A_i \cap W|}$$

The intuition behind this is that a voter gets diminishingly happy the more of her approved candidates are selected, so we give her a score of 1 for the first approved candidate selected, $\frac{1}{2}$ for the second approved candidate selected, and so on.

Remark 6.2.1.10. PAV is clearly infeasible to compute, we can construct a greedy version of it that is computable in polynomial time, though it of course, does not give the same outcome as PAV.

Definition 6.2.1.11 (Sequential Proportional Approval Voting (seqPAV))**.** We iteratively construct the committee by at each step adding a candidate that maximises the increase in the PAV score of the committee.

Algorithm 2 Sequential PAV

$W \leftarrow \emptyset$
while $|W| < k$ **do**
  Select $c^* \in C \setminus W$ maximizing score$(W \cup \{c^*\})$
  $W \leftarrow W \cup \{c^*\}$
end while
return W

Definition 6.2.1.12 (Generalized PAV ($\vec{w}$-PAV))**.** Use PAV with an arbitrary *weight vector* $\vec{w} = (w_1, w_2, \dots)$ where w_j is the weight given to a voter for having j approved candidates selected in the committee. Thus, we maximise

$$\text{score}_{\vec{w}}(W) = \sum_{i \in N} \text{score}_{\vec{w}}(i, W)$$

Where here $\text{score}_{\vec{w}}(i, W) = w_1 + w_2 + \cdots + w_{|A_i \cap W|}$.

Remark 6.2.1.13. If we consider the weight vector $\vec{w} = (1, 1, 1, \dots)$, then $\vec{w}$-PAV is nothing but Approval Voting.

For the vector $\vec{w} = (1, 0, 0, \dots)$, we get exactly Greedy-AV.

Can we make a similar comment about SAV? Think, why or why not?

6.2.2 Representative Committees

Idea 6.2.2.1. How do we formalise that a committee is representative of a preference profile?

Some intuition gives that groupings of size n/k should *have* at least one representative candidate (where n is the number of voters and k is the committee size, and n/k is an analog of the Hare quota in the apportionment setting).

A few problems with this intuition are:

- How do we group the voters into blocks of size n/k?
- What does it mean for a group of voters to have a representative candidate?

A first attempt could be the following. For any $S \subseteq N$ with $|S| \geq n/k$, there should be some candidate $c \in W$ *representing* S — but as mentioned above, what does it mean for c to represent S?

Instead, lets refine this approach. Call a voter group $S \subseteq N$ *cohesive* if there is some candidate c that all voters in S approve of, i.e. $\bigcap_{i \in S} A_i \neq \emptyset$. Then, we can say that a committee is representative if for every cohesive group S with $|S| \geq n/k$, there is some candidate in the committee that represents S, i.e. $\bigcap_{i \in S} A_i \cap W \neq \emptyset$. Where here, by represents we mean a candidate approved by all members in the cohesive voter group.

Example 6.2.2.2. Unfortunately, even this second approach does not quite work. It may be too demanding for any committee to satisfy.

Consider the example with $n = 4$ and $k = 2$, and the profile where: one voter approves of a only, the second voter approves of a and b, the third voter approves of b and c, and the fourth voter approves of c only.

$n/k = 2$ and voters (1, 2), (2, 3), and (3, 4) are cohesive groups of size n/k, each demanding that a, b, and c respectively be in the committee, but we can only select two candidates.

Definition 6.2.2.3 (Justified Representation (JR))**.** A committee W satisfies *justified representation* if for each cohesive group S with $|S| \geq n/k$, we have $\bigcup_{i \in S} A_i \cap W \neq \emptyset$.

Remark 6.2.2.4. This feels a bit strange, since we dont require that there is a candidate approved by all members of the cohesive group in the committee, but instead just a candidate approved by at least one member of the cohesive group. It is a very weak sense of representation.

An equivalent formulation makes a little more sense intuively. JR says that there does not exists a cohesive group of size $\geq n/k$ which is completely unrepresented.

Example 6.2.2.5. For $n = 12, k = 4$.

$$\text{Profile:} \qquad 3 \times \{a, b, c\}, 3 \times \{b\}, 3 \times \{e, f\}, 2 \times \{x, y, z\}, 1 \times \{x, y\}$$

Then JR enforces that b using the 3-sized cohesive group, and also that one of e, f and one of x, y, z are chosen.

Example 6.2.2.6. Despite JR being incredibly weak, many approval-based multiwinner rules violate JR.

- AV fails JR (for $k \geq 2$)
- SAV fails JR (for $k \geq 2$)
- seqPAV fails JR (for $k \geq 6$, found by linear programming)
- Greedy-AV satisfies JR (Exercise.)

Proposition 6.2.2.7 (Proof Idea)**.** *If there is a JR violation, then any candidate witnessing the violation can be swapped into the committee to increase the PAV score (by at least n/k).*

Theorem 6.2.2.8. *Assuming $w_1 = 1$, $w - PAV$ satisfies JR if and only if $w_j \leq \frac{1}{j}$ for all $j > 1$.*

Remark 6.2.2.9. Note that JR does not require that massive cohesive groups are represented more than once. For $\tau \in \mathbb{N}$, a group of size $|S| \geq \tau \cdot n/k$ *deserves* at least τ representatives.

Definition 6.2.2.10 (Extended Justified Representation (EJR))**.** Call a voter group $S \subseteq N$ *τ-cohesive* if there are at least τ candidates that all voters in S approve of, i.e. $|\bigcap_{i \in S} A_i| \geq \tau$.

A committee W satisfies *extended justified representation* if for each τ-cohesive group S with $|S| \geq \tau \cdot n/k$, we have,

$$|A_i \cap W| \geq \tau \qquad \text{For some } i \in S$$

Remark 6.2.2.11. It is clear that EJR imples JR, that is EJR is a stronger notion than JR. Thus, all those rules discussed in Example 6.2.2.6 that fail JR also fail EJR.

Example 6.2.2.12. Further, we see that Greedy-AV fails EJR (despite satisfying JR). Consider the example for $k = 3$ and $n = 100$,

$$98 \times \{a, b\}, \qquad 1 \times \{c\}, \qquad 1 \times \{d\}$$

The 98 $\{a, b\}$ voters form a 2-cohesive set, and form a cohesive set of size 98 which is greater than $2 \cdot n/k$, so they deserve at least 2 representatives, but Greedy-AV will select a (or b), then c, and then d.

Proposition 6.2.2.13. *PAV satisfies EJR (and thus JR too)*

Proof. Exercise. Try just show JR first, argue that a swap can improve the PAV score if there is an *unsatisfied* group. □

Theorem 6.2.2.14. *For any weight vector $\vec{w} \neq (1, 1/2, 1/3, \dots)$, $\vec{w}$-PAV fails EJR. (Assuming $w_1 = 1$).*

Proof. Omitted. □

Proposition 6.2.2.15. *It can be checked in polynomial time whether a given committee satisfies JR.*

In principle, this seems like a rather hard task, since we need to check for all subsets of voters.

Sketch. We check candidate by candidate instead. Iterate over each candidate, and check how many voter there are for it. If there are enough voter (i.e. $\geq n/k$), then we can consider if that candidate is included in the committee, if so we are happy and move on, if at any point we find a violation, then that gives witness to a committee not satisfying JR. □

Remark 6.2.2.16. Checking EJR is much harder, it is coNP-complete. We can verify an no instance by exhibiting a τ-cohesive group that is not sufficiently represented.

Idea 6.2.2.17. We now go to another axiom that is stronger than EJR but can be checked more efficiently.

Definition 6.2.2.18 (EJR$^+$)**.** A committee W satisfies EJR$^+$ if there does not exists, $c \in C \setminus W$, $S \subseteq N$, $\tau \in \mathbb{N}$, such that $|S| \geq \tau \frac{n}{k}$, and $c \in \bigcap_{i \in S} A_i$ and $|A_i \cap W| < \tau$ for all $i \in S$.

Intuitively, if a group S of size at least $\tau \cdot \frac{n}{k}$ and all members approve some common candidate $c \notin W$, then at least one member of S must have τ or more approved candidates in W.

Remark 6.2.2.19. Note the definition of EJR$^+$ does not require S to be τ-cohesive. This is the main difference from the notion of EJR.

Proposition 6.2.2.20. *EJR$^+$ implies EJR.*

Proof. Exercise.

□

Remark 6.2.2.21. It can be verified efficiently if a committee satisfies EJR$^+$.

In sketch, we iterate over candidates in $c \in C \setminus W$ and $l \in \{1, \ldots, k\}$ and check that the set of all voters approving c has the desired properties.

Intuition 6.2.2.22. In brief, we summarize the differences between EJR and EJR$^+$ in the following way.

EJR rules out groups S with

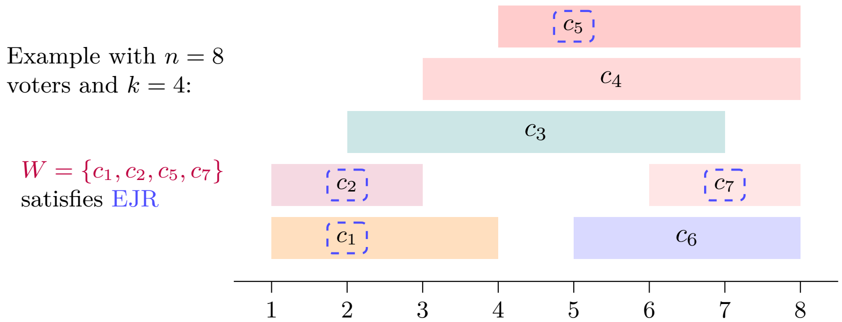
- $|S| \geq \ell \dfrac{n}{k}$
- $|A_i \cap W| < \ell$ for all $i \in S$
- $\left| \bigcap_{i \in S} A_i \right| \geq \ell$

EJR$^+$ rules out groups S with

- $|S| \geq \ell \dfrac{n}{k}$
- $|A_i \cap W| < \ell$ for all $i \in S$
- $\exists c \in \bigcap_{i \in S} A_i$ with $c \notin W$

Example 6.2.2.23. Consider the following example, the committee W indeed satisfies EJR (check), and yet that committee does not satisfy EJR$^+$, to see why, consider that c_4 is approved by all of the group $S := \{3, 4, 5, 6, 7, 8\}$, and is not in W.

Further $|S| \geq 3 \cdot n/k$, but every member is satisfied by fewer than 3 candidates in W.



Remark 6.2.2.24 (Finding Proportional Committees Efficiently)*.* So far, we have largely talked about verifying a committee satisfies certain properties, but we have not talked about how to find a committee that satisfies these properties.

- A committee satisfying JR can be found efficiently (Exercise: Why?)
- PAV satisfies EJR$^+$ (and EJR) but as noted, cannot be computed efficiently

Theorem 6.2.2.25. *A committee satisfying EJR$^+$ can be computed efficiently.*

Sketch. Start with an arbitrary committee of appropriate size. Iterate over all pairs of candidates where one candidate is in the committee and one is outside of the committee. Determine if swapping them would increase the PAV score of the committee, if so we swap them and repeat. Eventually, we reach some *PAV-Score local minima*, and then we terminate.

It turns out, that this local minima must satisfy EJR$^+$. □

Remark 6.2.2.26*.* For several reasons, this doesn't make for a particularly convincing or *principled/nice* voting rules. Later, people came up with sequential rules that also satisfy EJR$^+$, and are also polynomial time computable.

Definition 6.2.2.27 (Profitable Deviation)**.** Given a committee W and a group of voters $S \subseteq N$, a set of candidates $T \subseteq C$ is a *profitable deviation* if

$$|T \cap A_i| > |W \cap A_i| \qquad \text{for all } i \in S$$

Definition 6.2.2.28 (Core Stability)**.** A committee W is *core stable* if there is no group of voters S and set of candidates T such that $|T| \leq |S| \cdot \frac{k}{n}$ and T is a profitable deviation for S from W.

The *core* consists of all committees that are core stable.

Remark 6.2.2.29*.* It is an open question whether the core is always non-empty.

Proposition 6.2.2.30 (EJR's relation to Core Stability)**.** *Any core stable committee is EJR, that is core stability is a stronger property than EJR.*

Idea. Show the contrapositive. □

Example 6.2.2.31. We may have that a committee is EJR but not core stable.

For example, with $k = 10$, $n = 20$ and $C = \{x_1, \ldots, x_{10}, y, z\}$,

$$3 \times \{x_1, y\} \qquad 3 \times \{x_1, z\} \qquad 14 \times \{x_2, \ldots x_{10}\}$$

$W := \{x_1, \ldots, x_10\}$ satisfies EJR, but the group of voters approving y and z can profitably deviate to $\{x_1, y, z\}$.

This exemplifies the difference between the notions. Core stability considers all possible deviations, whereas EJR only considers deviations restricted to cohesive groups.

6.2.3 Approval-based Multiwinner Voting's relation to Apportionment

Idea 6.2.3.1. The apportionment setting may be viewed as a special case of the approval-based multiwinner voting setting, where we partition the candidates into sets $C_1, \ldots, C_m$ (where $C_j \geq k$ [2]), and each voter $i \in N$, has $A_i = C_j$ for some j.

In this setting, $v_j = |N_j|$ where $N_j = \{i \in N : A_i = C_j\}$, and $k = h$.

Thus, we can see the apportionment setting as a special case of the approval-based multiwinner voting setting, and can consider how our approval-based multiwinner rules perform in this setting.

Example 6.2.3.2 (Apportionment to Multiwinner Translation)**. Step 1:** Translate $(v, h) \longrightarrow (A_N, k)$.

Given $v = (30, 18, 7)$, $h = 4$. Create $|C_j| = k = 4$ candidates per party:

$$C = \underbrace{\{a_1, a_2, a_3, a_4\}}_{P_1} \cup \underbrace{\{b_1, b_2, b_3, b_4\}}_{P_2} \cup \underbrace{\{c_1, c_2, c_3, c_4\}}_{P_3}$$

Set $k = h = 4$. The approval profile is:

$$\begin{array}{rl} 30\times & \{a_1, a_2, a_3, a_4\} \\ 18\times & \{b_1, b_2, b_3, b_4\} \\ 7\times & \{c_1, c_2, c_3, c_4\} \end{array}$$

Step 2: Apply rule to (A_N, k): say, $W = \{a_1, a_2, a_3, b_1\}$.

Step 3: Read off seat distribution: $(|W \cap C_1|, |W \cap C_2|, |W \cap C_3|) = (3, 1, 0)$.

Theorem 6.2.3.3 ($\vec{w}$-PAV induces Divisor methods)**.** *Let $\vec{w} = (w_1, w_2, \ldots)$ be a weight vector with $w_j \geq w_{j+1} > 0$ for all $j \in \mathbb{N}$. Then, the apportionment method induced by w-PAV coincides with the divisor method based on divisor sequence $d(j) = 1/w_j$.*

- *PAV (weight vector $w_j = 1/j$) induces the D'Hondt method (divisors $1, 2, 3, \ldots$).*
- *w-PAV with $w = (1, 1/3, 1/5, \ldots)$ induces the Sainte-Laguë method (divisors $1, 3, 5, \ldots$).*

Proposition 6.2.3.4. *If an approval-based multiwinner rule satisfies EJR, then the apportionment method it induces satisfies the lower quota property.*

Observation 6.2.3.5. Recall that DHondt's method is the only divisor method satisfying the lower quota property, and likewise PAV is the only $\vec{w}$-PAV rule satisfying EJR.

[2]To ensure that, in principle, each party could fill all seats.

Bibliography

[1] Felix Brandt, Vincent Conitzer, Ulle Endriss, Jérôme Lang, and Ariel D. Procaccia, editors. *Handbook of Computational Social Choice.* Cambridge University Press, 2016.

[2] Edith Elkind, Martin Lackner, and Dominik Peters. Preference restrictions in computational social choice: A survey. *CoRR*, abs/2205.09092, 2022.